OECD Statistics Working Papers 2025/01

Towards more timely measures of labour productivity growth

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https://dx.doi.org/10.1787/436ecbb5-en

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Abstract

Productivity developments have become increasingly uncertain in recent years due to a series of shocks, including the COVID-19 pandemic, the energy crisis, and rising geopolitical tensions. Despite this, attempts to nowcast recent trends in productivity growth have been limited, often focusing on micro-level productivity within specific occupations or industries. To date, no effort has been made to nowcast macroeconomic measures of labour productivity growth across a broad group of countries – a gap this paper seeks to address.
This paper presents nowcast of labour productivity growth over a panel of 40 OECD and accession countries. A key novelty of this paper is the integration of machine learning techniques with mixed-frequency models within a panel framework, enabling the optimal utilisation of higher-frequency data. The approach combines mixed-frequency setups with a diverse range of models, including dynamic factor models, penalised regressions (LASSO, Ridge, ElasticNet) and tree-based models (Gradient Boosted Trees, Random Forests) and accounts for publication lags. Performance gains compared to an autoregressive benchmark average around 35% across the 40 countries. Machine learning models, in particular Gradient Boosted Trees, are found to outperform alternatives in most countries. A MIDAS specification with estimated weight is found to bring additional information compared to an approach with imposed weights in 30 out of 40 countries.

Keywords: Nowcasting, machine learning, panels, labour productivity.

JEL codes: C4, C53, C23, E24.

Résumé

L'évolution de la productivité est devenue de plus en plus incertaine ces dernières années en raison d'une série de chocs, notamment la pandémie de COVID-19, la crise énergétique et la montée des tensions géopolitiques. Malgré cela, les tentatives de prévision immédiate (nowcasting) des tendances récentes de la croissance de la productivité ont été limitées, se concentrant souvent sur la productivité à l'échelle microéconomique, dans des professions ou des industries spécifiques. À ce jour, aucun effort n'a été entrepris pour prévoir à court terme les mesures macroéconomiques de la croissance de la productivité du travail à travers un large groupe de pays, une lacune que cet article vise à combler.
Cet article propose des prévisions immédiates de la croissance de la productivité du travail sur un panel de 40 pays membres de l'OCDE et pays candidats. Une nouveauté majeure de ce travail réside dans l'intégration dans le cadre d'un panel de techniques de machine learning avec des modèles à fréquence mixte, permettant une utilisation optimale des données à haute-fréquence. L'approche combine des configurations à fréquence mixte avec une diversité de modèles, y compris des modèles à facteurs dynamiques, des techniques de régression pénalisées (LASSO, Ridge, ElasticNet) et des modèles basés sur des arbres (Gradient Boosted Trees, Random Forests) et prend en compte les délais de publication.
Les gains de performance, par rapport à un modèle autorégressif, atteignent en moyenne près de 35 % sur l'ensemble des 40 pays. Les modèles de machine learning, en particulier les Gradient Boosted Trees, surpassent les autres approches dans la majorité des pays. Une spécification MIDAS (Mixed Data Sampling) avec des poids estimés apporte des informations supplémentaires par rapport à une approche avec des poids imposés dans 32 des 40 pays.

Mots-clés : Nowcasting, machine learning, pannel, productivité du travail.

Codes JEL: C4, C53, C23, E24.

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Towards more timely measures of labour productivity growth

Yann-Yves Dorville, Nhung Luu, Annabelle Mourougane and Julia Schmidt[1]

1. Introduction

1. Productivity is a key determinant of long-term growth and closely monitored by policy analysts. It is used as a diagnosis tool to examine economic developments and help target the right policy to boost economic performance, including international competitiveness but also long-term growth and living standards (Krugman, 1994). It also helps to monitor the fallout of an economic shock or the impact of reform. As such, analysts often look for timely productivity measures.

2. Annual labour productivity indicators are regularly published in the OECD Productivity database and in the OECD Compendium of Productivity Indicators. However, official data are usually released with a lag of one or two years. While in theory quarterly GDP and hours worked data can provide insights on recent productivity developments, but those data are not available for all OECD and accession countries. When those data exist, they are subject to sometimes significant revisions, especially during exceptional periods such as the COVID-19 pandemic.

3. Nowcasting, which collapses in one term 'now' and 'forecasting', refers to the prediction of economic subjects in current time, the very near future and the very recent past. The basic principle of nowcasting is to exploit the information which is published early and possibly at higher frequencies than the variable of interest to obtain an 'early estimate' before the official figure becomes available (Bánbura et al., 2013). Nowcasting techniques are widely employed to predict GDP growth (Dauphin et al., 2022; Fosten and Greenaway-McGrevy, 2022), inflation (Modugno, 2012; Macias, Stelmasiak and Szafranel, 2022) or trade (Martines-Martin and Rusticelli, 2020; Jaax, Gonzales and Mourougane, 2024; Dubovik et al., 2022; Chinn, Meunier and Stumpner, 2023; Hopp, 2022.)

4. Attempts to nowcast most recent developments in productivity growth have been scarce, focusing very often on micro-level productivity of specific occupations or industries (e.g. Goodarzizad et al. (2023) for the construction sector, Tam et al. (2002) for excavation, Hanna et al. (1999) for electrical construction). However, to our knowledge, there has been no attempt to nowcast macro-economic measures of productivity growth for a large group of countries, which is the focus of this paper.

5. Labour productivity, measured in terms of employment or of total hours worked, and multifactor productivity are the most common indicators of productivity. Those different concepts respond to different

[1] The work has benefited from the help of Itai Walk. The authors are all from the OECD Statistics and Data Directorate. They would like to thank Steve MacFeely (OECD Statistics and Data Directorate), Alexander Jaax (OECD Trade and Agriculture Directorate), Christophe André, Martin Borowiecki, Pierre-Alain Pionnier, Elena Rusticelli (all from the OECD Economics Department), and participants from the INSEE-OECD workshop on times series (December 2024). Special thanks to Virginie Elgrably for excellent technical assistance.

demand. In theory, labour productivity measured in terms of hours worked is a neater concept than labour productivity measured in terms of employment. Multifactor productivity is usually computed as the Solow residual of a production function and is by nature non observable and more challenging to nowcast.

6. Against this background, this paper seeks to nowcast annual labour productivity growth, measured in terms of GDP per hours worked over a sample of 40 OECD and accession countries. The novelty of the approach lies in its combination of mixed-frequency set ups with a range of models, including Dynamic Factor Models (DFM), penalised regression models (LASSO, Ridge, Elastic Net) and tree-based models (Gradient Boosted Trees (GBM), Random Forest). All are applied in a panel setting to cope with limited data availability at the country level. In addition, mixed-data sampling (MIDAS) allows to make use of higher frequency information in an optimal way. There have been only a few examples combining machine learning and MIDAS in the literature (Babii et al., 2023; Borup et al., 2022).

7. Features used to nowcast are selected using the body of evidence on the drivers of productivity, in particular OECD/APO (2023). Analysis of the drivers of productivity, including most recently Artificial Intelligence, has been extensively surveyed within the OECD and elsewhere and is out of the scope of this paper. Those analyses are however accounted for in the choices of features used to nowcast productivity growth.

8. The main findings from the empirical analysis are as follows:

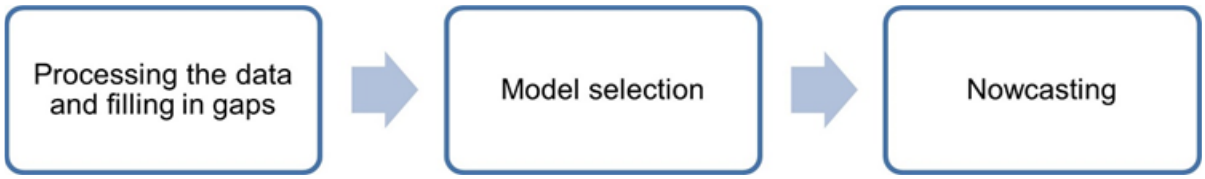
- Selected model performance appears to be satisfactory, with an average error of less than 10% for the one-year-ahead prediction of labour productivity growth on average over the period 2015-22 and across OECD countries. The models do not fully capture the un-usual volatility in labour productivity growth that occurred during the COVID-19 crisis, but the magnitude of errors is low relative to the extent of the crisis and its implications on growth and labour markets.
- Performance gains compared to an autoregressive benchmark average around 35% across the 40 countries. Performance appears to be better for countries exhibiting lower productivity variability. Gains are close or above 60% for Denmark, France, Romania, Latvia, and Slovenia.
- Machine learning models, in particular GBM, are found to outperform alternatives in most countries. A MIDAS specification with estimated weight is found to bring additional information compared to an approach with imposed weights in 30 out of 40 countries, 75% of the cases. The difference in predictive accuracy is significant according to a Diebold-Mariano test.
- Nowcasts point to labour productivity growth of about 0.9% in 2023 on average across OECD countries (excluding Türkiye), close to the average 2001-19). It would rebound to 1.8% in 2024, but a less robust recovery cannot be ruled out. Estimates are surrounded by large uncertainties, especially in countries such as Ireland, Costa Rica and Colombia.

9. The remainder of the paper is structured as follows: Section 2 provides an overview of the approach, focusing on its main novelties. Section 3 presents the data. Section 4 focuses on model selection. Section 5 describes productivity developments in 2023-24 as foreseen by the best selected models. A last section concludes. A literature review and technical details are provided in Annex A and Annex B and additional results are presented in the following Annexes.

2. Overview of the empirical strategy

10. This section provides a short overview of the approach to nowcast labour productivity growth, measured in terms of total hours worked, for 40 OECD and accession countries (Figure 1).

Figure 1. Main steps of the empirical strategy



Source: OECD illustration.

11. The first step is to collect and process features of annual, quarterly, and monthly frequency that are used to predict labour productivity growth (the target variable). Those include national accounts data, labour market indicators, trade and business statistics, as well as measures of geopolitical and financial risks and uncertainty. Indicators, which are not sufficiently timely, have been prolonged using machine learning models or ARIMA models in the case of monthly indicators.

12. A key specificity of the approach is to run models in a panel setting. As the target variable is annual and only available for the years 1995-2022, a panel approach increases the robustness of results (Woloszko, 2020; Fosten and Nandi, 2023; Mourougane et al., 2023). In addition, models are estimated in a quasi real-time accounting for publication lags, identified using lags observed in 2023.

13. The panel setting was combined to a mixed-data sampling (MIDAS) approach, initially developed by Ghysels et al. (2004) (see Annex B). MIDAS allows to use the information contained in high-frequency series to predict lower-frequency variables in an efficient way. Several approaches were tested:

- The unrestricted MIDAS (U-MIDAS) approach, in which each higher-frequency variable enters the model independently with its own coefficient.
- The restricted MIDAS (R-MIDAS) approach, which imposes a lag-polynomial structure on the higher-frequency data, reducing the number of parameters that need to be estimated.
- An imposed-weight MIDAS approach (I-MIDAS), which is a special case of the R-MIDAS, with equal weights for each higher-frequency variable.

14. The second step is to identify the best model to nowcast labour productivity growth. A range of models are evaluated, including Dynamic Factor Models (DFM), penalised regressions (LASSO, Ridge and Elastic Net), tree-based approaches such as random forests and gradient boosted trees (GBM). Consistent with the nowcasting literature, the performance of all these models is compared to that of an AR(1) model, given the annual nature of the data. Choosing another benchmark would not modify the ranking and thus the choice of the best selected models (Annex E). Ultimately, 19 combinations were tested (6 model types (e.g. GBM) * 3 approaches (restricted or unrestricted MIDAS, imposed-weights MIDAS) +1 benchmark). In addition, several versions of the machine-learning models are tested, to identify the best parametrisation of the model.

15. Best performing models are selected based on the Root Mean Squared Errors (RMSEs) for one-year-ahead predictions. The model that performs the best is selected for each country, i.e. one best model per country, rather than using the same model that would perform best on average across all the countries. Following Hastie et al. (2009), a cross-validation process is adopted to prevent overfitting – i.e. a situation when the model performs well in-sample, but fails in generalising out-of-sample. This involves splitting the data into training sets to learn the model parameters, and test sets, to evaluate the model's performance

on unseen data. Cross-validation is conducted using an expanding window approach from 2015 to 2022, with the window length set sufficiently large to ensure that enough observations are available to produce standard performance metrics. In the case of R-MIDAS, the Profile Likelihood method allows the joint estimation of model parameters and the polynomial structure to be estimated in an efficient way (Ghysels and Qian, 2018).

16. The last step applies the best models to nowcast productivity growth for 2023-24. Consistent with the economic literature a "consensus" model, which average the five best models, is also presented (Jaax et al., 2023; Kotchoni et al., 2019). The rationale is that the average smoothes the nowcast and lowers the impact of potential feature outliers. A confidence interval is constructed around the estimates to reflect the degree of uncertainty using model-specific past RMSE as a proxy of the point estimate expected standard deviation. Finally for methods where this was possible relative feature importances were reported to provide information on the factors explaining the nowcasts.

3. Data

17. Labour productivity growth, the target variable, is sourced from the OECD Productivity database. Explanatory variables are of varying frequencies and seek to capture key economic drivers of productivity (see Table 1 for some examples). In total, the dataset contains 47 series per country, over the period 1995-22 (147 when considering different frequencies of high-frequency variables as distinct features, as is done in the unrestricted version of MIDAS). In addition, global indicators, for instance of uncertainty, were also added. A complete list is provided in Annex D.

Table 1. Examples of explanatory variables

	Example of variables
Activity	GDP Gross output by industry Gross fixed capital formation Capacity utilisation
Labour market and prices	Wage Unemployment and employment Firm entry and turnover Consumer Price Index Producer Price Index
External environment	Openness to trade Exports and imports volumes and values FDI investment Global supply chain pressure index
Financial development and uncertainty	Geopolitical risk index World uncertainty index Stock market volatility

Note: Most variables are extracted from OECD databases. See Annex D for a complete list of variables and their sources.
Source: Authors' compilation.

18. Data pre-processing was conducted prior to model training and testing, with some specific steps to fit the requirements of the individual methods. First, missing data in the period 2015-22 were imputed with a special attention to prevent data leakage. For variables with missingness occurring at any point within the series, the missing data were imputed using the median value of the available data point from that series. In cases where a variable was completely missing, data from a neighbouring country or, alternatively, from the region, were used. Imputation using mean value could be an alternative. However,

this choice is prone to bias, particularly in small samples or when the data includes outliers. Details on the imputation methods can be found in Annex D.

19. Second, monthly and quarterly series were seasonally adjusted. All series were normalised using min-max scaling, ensuring that all features are on a comparable scale. This improves model performance by preventing any single feature from dominating the learning process due to its scale, while also reducing the time needed to estimate models (de Amorim et al., 2023). Normalisation was applied independently to each training set to prevent data leakage from the test set into the training phase.

20. Third, variables which did not pass the Augmented Dickey-Fuller stationary test were transformed through first differencing to achieve stationarity.

21. Fourth, to prevent a high degree of multicollinearity, which would negatively impact the predictive accuracy of penalised regression models (LASSO, Elasticnet and Ridge) and the DFM, highly correlated variables were removed through an iterative process. In the case of the DFM, explanatory variables were first ranked in descending order based on the amount of missing data, with those having the most missing values ranked highest. Next, starting with the top-ranked variable, each variable was regressed against the remaining variables to calculate the variance inflation factor (VIF) score. A variable was eliminated it its VIF exceeded the threshold of ten. This ranking by missingness ensures that variables with lower data quality are removed first if they exceed the VIF threshold. Consequently, subsequent VIF calculations exclude variables that have already been dropped, ensuring a focus on variables with higher data quality.

4. Model performance

22. For each country, the best nowcasting model was selected based on its one-year-ahead predictive performance. Model performance is assessed using the relative Root Mean Squared Error (RMSE), a metric that compares the average prediction errors of a model to a benchmark autoregressive model of order one over the test sample. A relative RMSE below one indicates that the model outperforms the benchmark. In addition, the Harvey-Leybourne-Newbold test, a variant of the Diebold Mariano test for small samples, helps to determine whether the predictive accuracy differences between selected models and the benchmark are statistically significant (Harvey, Leybourne and Newbold, 1997). Additional metrics have been calculated, such as the two-year ahead predictive performance and the Forecasting Directional Accuracy (FDA) which measures how often the predicted direction of growth (up or down) matched the actual direction. Those metrics are however not used in the selection process.

Machine learning models, in particular Gradient Boosted Trees are found to outperform alternatives

23. Model performance appears to be overall satisfactory at the OECD economy wide level. The average error was less than 10% for the one-year-ahead prediction of labour productivity growth on average over the period 2015-22 and across OECD countries. The “best” nowcasting models tend to perform better than the AR(1) benchmark when performance is measured in terms of one-year ahead RMSE (Table 2). The benchmark is outperformed in all the countries, but Colombia. A comparison of model performance using DFM as an alternative benchmark is provided in Table A E.2. Performance relative to the Dynamic factor Model in Annex E.

24. Performance gains compared to the autoregressive benchmark average almost 35% across the 40 countries (Figure 2, Panel B). They would amount to 43.4% if a DFM was used as a benchmark. Model performance appears to be better for countries exhibiting lower productivity variability. Accuracy gains are especially important for Denmark, France, Romania, and Slovenia with gains close or above 60% (Figure 2, Panel A).

Table 2. Best model selection based on one-year ahead RMSE

Models							
	Benchmark AR (1)	LASSO	Ridge	Elastic Net	Random Forest	Gradient Boosted Trees	Dynamic Factor Model
Number of instances selected as best model	1	0	3	0	0	**35**	1

MIDAS methods			
	Imposed-weight MIDAS	Un-restricted MIDAS	Restricted MIDAS
Number of instances selected as best model	9	19	11

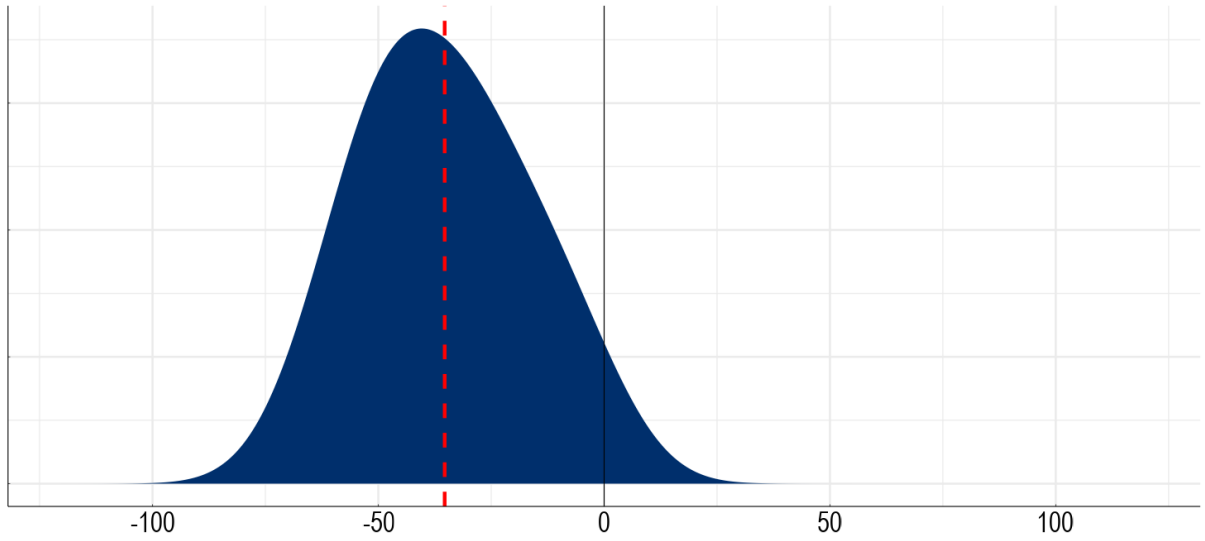
Note: Information on the nowcasting models can be found in Section 2 and Annex B.
Source: Authors' calculation.

25. The FDA suggests that nowcasting models manage to predict the direction of annual developments in labour productivity growth in most cases. Indeed, for 40 countries, the share of one-year ahead models for which actual and predicted annual changes have the same sign is found to be on average around 85%. FDA is maximal in 13 countries, which suggests very good performance of the model in detecting the direction of change. FDA is below average in Austria, Mexico, Spain, and the Netherlands (62.5%), and even more so in Luxembourg and Portugal (50%). In these countries, however, the magnitude of nowcasting errors is relatively small.

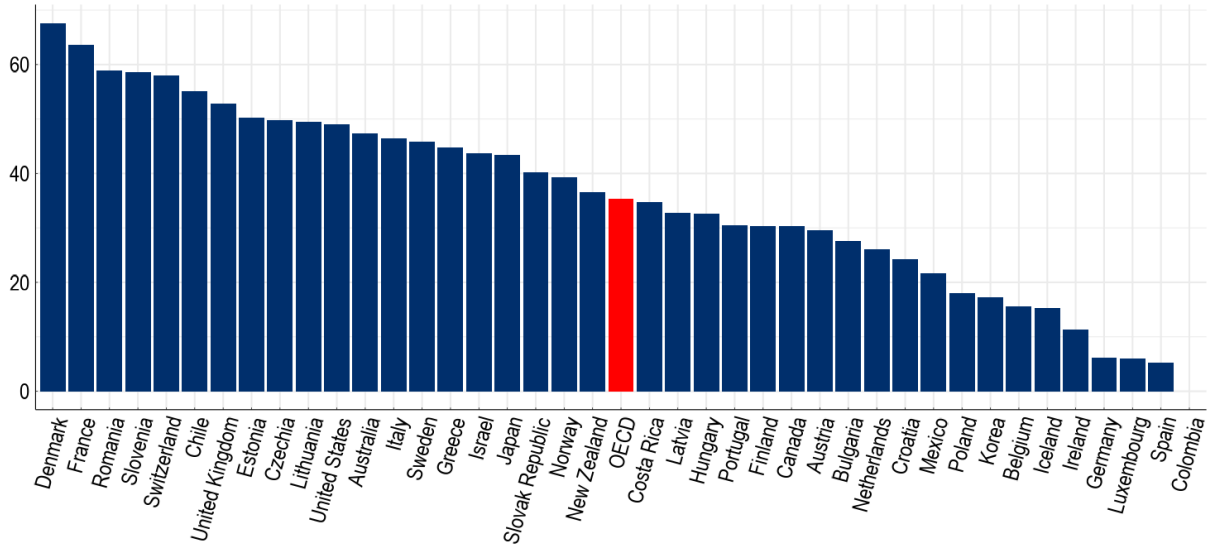
26. The GBM is selected as the best model in all countries but Canada, Ireland, Lithuania, Columbia, and Spain, indicating that it has, on average, a higher predictive accuracy than the other models. In Canada, Lithuania and Ireland, Ridge is found to be the best performing model, while the DFM is selected for Spain. No model outperforms the AR for Colombia. Surprisingly, penalised regressions models appear to perform poorly. The prevalence of GBM is consistent with what other authors have found for similar types of exercise (Mourougane et al., 2023).

Figure 2. Accuracy gains by countries over the out-of-sample period (2015-22)

Panel A. Distribution of gains between best models and AR(1) across countries



Panel B. Accuracy gains (RMSE differences between best model and AR(1)), percentage



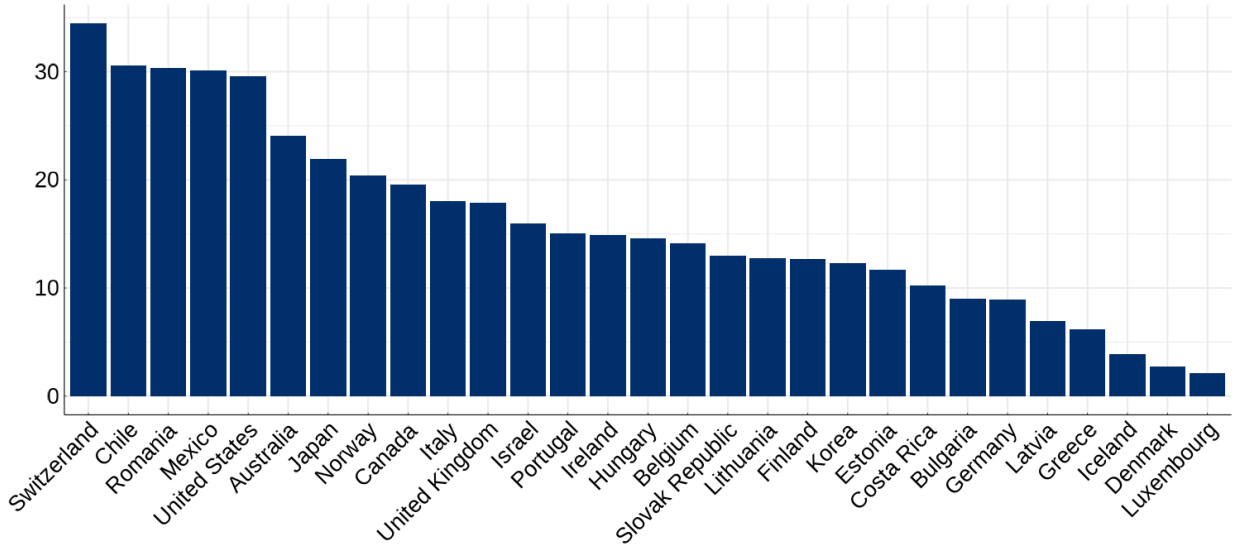
Note: The distribution in Panel A is estimated with a Kernel Density Estimation algorithm (Scott, 1992) with a Gaussian kernel and a bandwidth parameter of 1.5. The x-axis represents the accuracy gains (computed as reduction in RMSE in percent). Panel B shows the country-specific reduction in RMSE in absolute value. The red line represents the mean decrease in RMSE across countries.
Source: Authors' calculations.

The use of most flexible forms of MIDAS appears to be useful

27. At first glance, the use of the most flexible form of MIDAS does not seem to improve markedly nowcasting performance compared to the versions where weights are imposed. The type of model selected depends on the country specificity but not on the feature of the MIDAS framework. In particular, GBM is the most selected methods, in all three sampling frameworks considered.

Figure 3. Gains from using flexible forms of MIDAS

Percentage difference in RMSE between the best selected models and the best imposed-weight MIDAS model



Note: Models are estimated using the restricted MIDAS, the unrestricted MIDAS, and the imposed-weight MIDAS. The best selected model corresponds to the best model all methods included. Imposed-weight MIDAS refers to the conventional practice consisting of averaging high-frequency variables with uniform weights. A large difference in performance between imposed-weight and other methods of MIDAS is indicative of substantial additional information in high-frequency variables. This figure only includes countries for which said difference is positive.
Source: Authors' calculations.

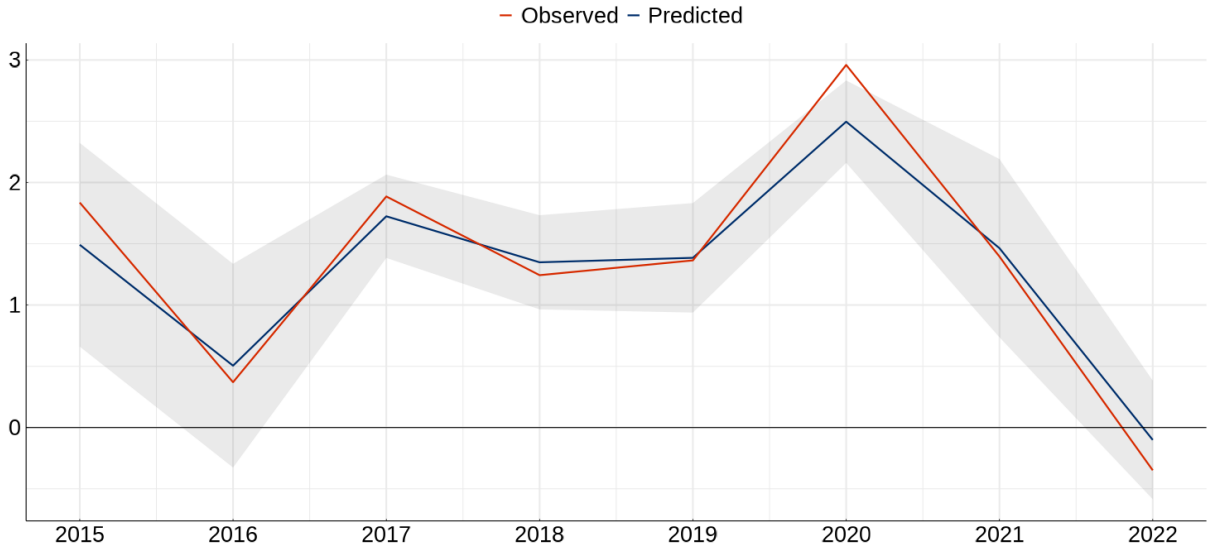
28. Although on average across countries the improvement in nowcasting performance resulting from the introduction of the most flexible form of MIDAS are small, a MIDAS specification with estimated weight is found to bring additional information compared to an approach where weights are imposed in 30 out of 40 countries, 75% of the cases. Unrestricted MIDAS accounts for 63.3% of such cases and restricted MIDAS for 36.7%. The difference in predictive accuracy is significant according to a Diebold-Mariano test. In addition, sizeable differences are visible for some specific countries (Figure 3). Gains, as defined by the percentage difference in RMSE between the best selected models and the best imposed-weight MIDAS model exceed 30% in four countries (Switzerland, Chile, Romania and Mexico) and 10% in 22 countries.

Performance varies over time

29. Performance varies over time, with no clear pattern at the aggregate level. The models do not fully capture the unusual volatility in labour productivity growth that occurred during the COVID-19 crisis (Figure 4). However, the RMSE for the average OECD countries would have been lowered by 12.5% only, if computed only over the pre-COVID period. This is relatively good, given that the crisis was triggered by non-economic factors while the nowcasting models tested here rely exclusively on economic indicators. In addition, specific policies that were put in place during that time may have distorted labour-market developments as job retention schemes were introduced in some European economies, leading to an artificial spike in productivity growth (see OECD, 2023; OECD/APO, 2022).

Figure 4. One-year ahead model performance across OECD countries

Labour productivity growth, percentage



Note: Observed data refers to the unweighted average labour productivity growth, predicted data corresponds to the unweighted average of predicted labour productivity growth. The shaded area displays the 95% confidence band associated to the point estimates for the OECD aggregate. They are computed as $\left[\hat{\mu} \pm p_{97.5}\, \hat{\sigma}/\sqrt{N}\right]$ where $\hat{\mu}$ corresponds to the unweighted average prediction and $\hat{\sigma}$ the average RMSE, $p_{97.5} \approx 1.95$ is the 97.5th percentile of the normal distribution, and $N = 40$ the number of countries considered. Türkiye is excluded from the sample due to a large share of missing input data.
Source: Authors' calculation.

Model performance appears to be negatively correlated with productivity variability

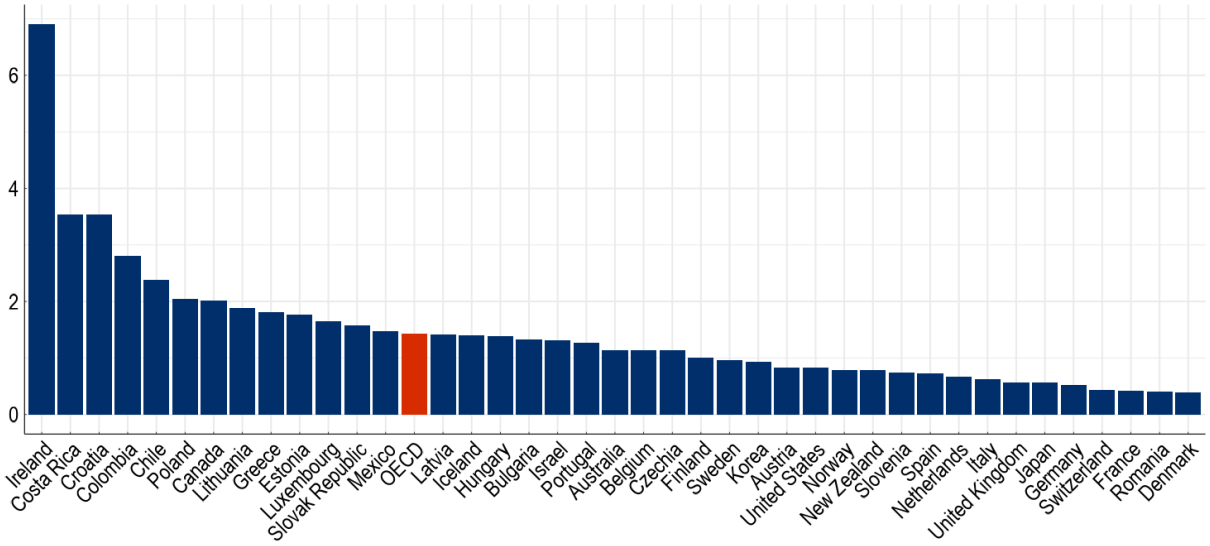
30. Good overall performance at the OECD-wide level masks a wide diversity of country-level model performance (Figure 5). In general, model performance in large economies is found to be better than in smaller, more volatile economies. In particular, the United States, Japan, Germany, Italy and France, and to a lesser extent the United Kingdom, exhibit lower absolute RMSEs (of one percentage point or less) than the average of countries. By contrast, poorer model performance can be observed in smaller OECD countries. Ireland stands out particularly, with deviations of 6.9 percentage points from the average country RMSE (at 1.4 percentage points). RMSEs are also relatively high for Costa Rica, Colombia and Croatia and to a lesser extent Chile.

31. The main source of error varies across countries:

- There is a relationship between the variability of the target variable and the magnitude of nowcasting errors (Figure 6). In particular, the volatility is very high in Ireland, and to a lesser extent Colombia, Costa Rica, Croatia and Chile. Those are the countries which exhibit the highest errors in sample.
- In a number of countries, a failure to capture the high volatility of labour productivity growth in the first years of the COVID-19 crisis explains most of this underperformance (Figure 7). This is in particular the case in some Latin America countries (Colombia and Costa Rica), but also for Croatia.
- Missing input data might also weigh on the prediction performance. For Chile, Canada, Croatia, Estonia and Ireland around 10% of observations across input features require imputation.

Figure 5. Absolute RMSEs are lower in large economies

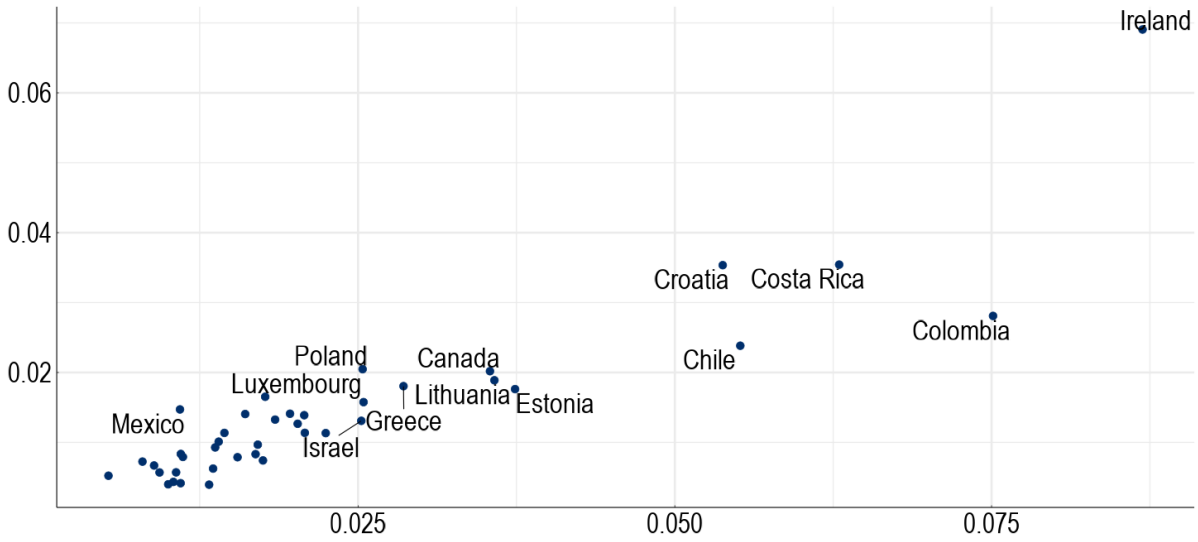
Average RMSE over 2015-22, percentage



Note: Country-average RMSE (Root Mean Squared Errors) were estimated across the pseudo-out-of-sample period. Predictions are obtained from selected best models and MIDAS methods (See Section 2 for a complete description). The red bar corresponds to the unweighted average RMSE over OECD countries.
Source: Author calculations.

Figure 6. Variability of labour productivity growth is correlated with nowcasting errors

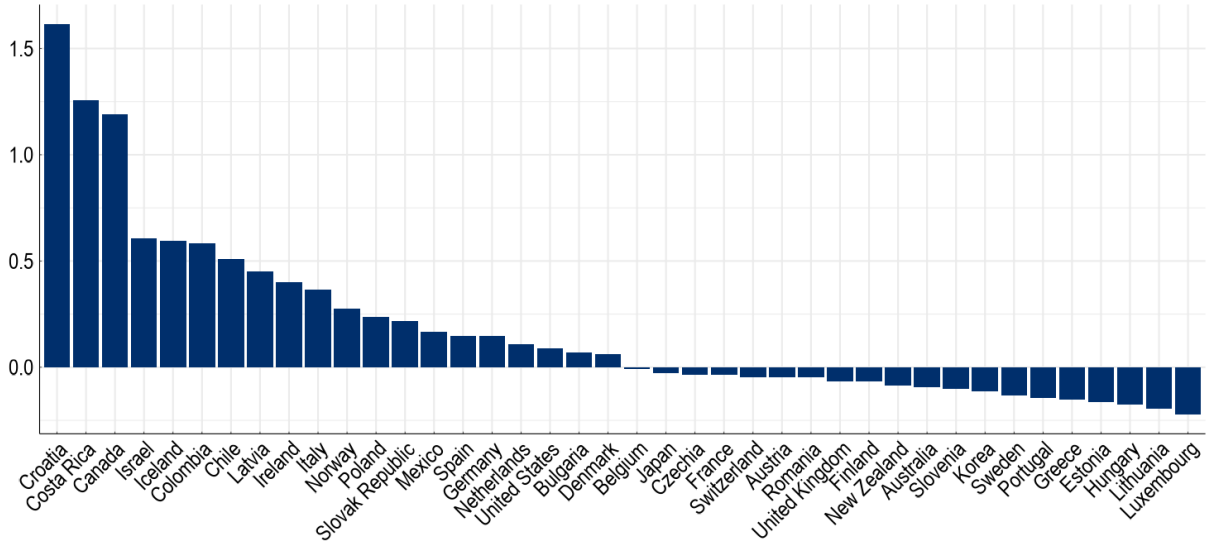
Standard deviation of labour productivity (x-axis) against out-of-sample RMSE (y-axis) over 2015-22



Note: Predictions are obtained from selected best models and MIDAS methods (See section 2 for a complete description).
Source: Authors' calculations.

Figure 7. Comparison of predictions errors over time

Change in RMSE over 2015-22 compared to the No-COVID period (2015-19, 2022), percentage points



Note: Predictions are obtained from selected best models and MIDAS methods (See section 2 for a complete description).
Source: Authors' calculations.

5. Nowcasting labour productivity developments in 2023-24

32. Nowcasts for the years 2023-24 are derived using the best selected models and are presented together with the average of the five best models (the so-called "consensus" model). Such a model is often found to perform well, because of its capacity to iron out the importance of outlier observations.

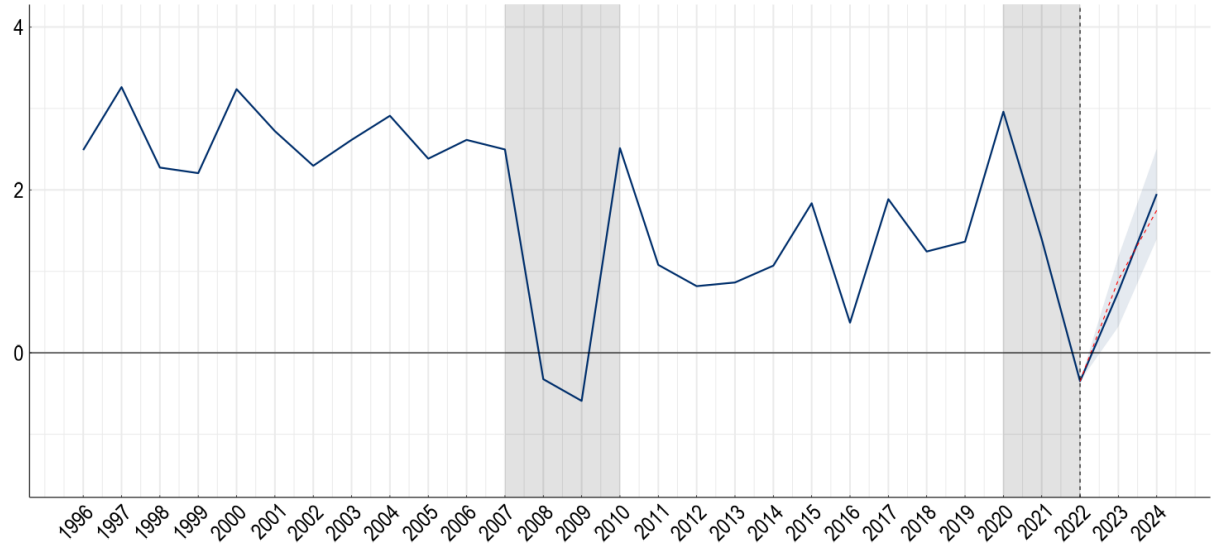
33. A qualitative reading of economic developments in 2023-24 points to a mixed picture. On the one hand, economic uncertainty and financial volatility decreased – although geopolitical risks clearly rose. On the other hand, labour markets were tight in many countries, but growth has remained subdued, especially in Europe. The resulting impact on productivity growth in 2023-24 remains hard to predict.

34. Using best performing models selected in the preceding sections, average labour productivity growth across OECD countries (excluding Türkiye) is estimated at 0.9% in 2023, close to pre-pandemic average of 2001-19, strengthening to 1.8% in 2024 (Figure 8). The consensus model points to very similar nowcasts. Although labour productivity growth seems to be relatively robust for 2024, the possibility of experiencing negative productivity growth cannot be ruled out in most countries. This suggests that the 2024 estimates should be interpreted with great caution.

35. De Vries (2023) and Aldama et al. (2023) also foresee a modest productivity growth in 2023 compared to 2022 and a more optimistic outlook for 2024 and beyond. The absence of a significant improvement in productivity growth in 2023 is consistent with the moderate real GDP growth experienced that year, coupled with a decrease in hours worked relative to that of 2022. A rebound in 2024 is also expected in the latest short-term projections of the Economic Outlook (OECD, 2024).

Figure 8. Labour productivity growth nowcast for 2023 and 2024

OECD average, percentage



Note: The blue shaded area displays the 95% confidence band associated to the point estimate for the OECD aggregate. They are computed as $\left[\hat{\mu} \pm p_{97.5}\,\hat{\sigma}/\sqrt{N}\right]$ where $\hat{\mu}$ corresponds to the unweighted average prediction and $\hat{\sigma}$ the average RMSE, $p_{97.5} \approx 1.95$ is the 97.5th percentile of the normal distribution, and $N = 40$ the number of countries in the respective region. The red dotted line depicts the consensus nowcast. Grey shaded areas represent global crisis episodes. Türkiye is excluded from the sample due to a large share of missing input data. Predictions are obtained from selected best models (See section 2 for a complete description).
Source: Authors' calculation.

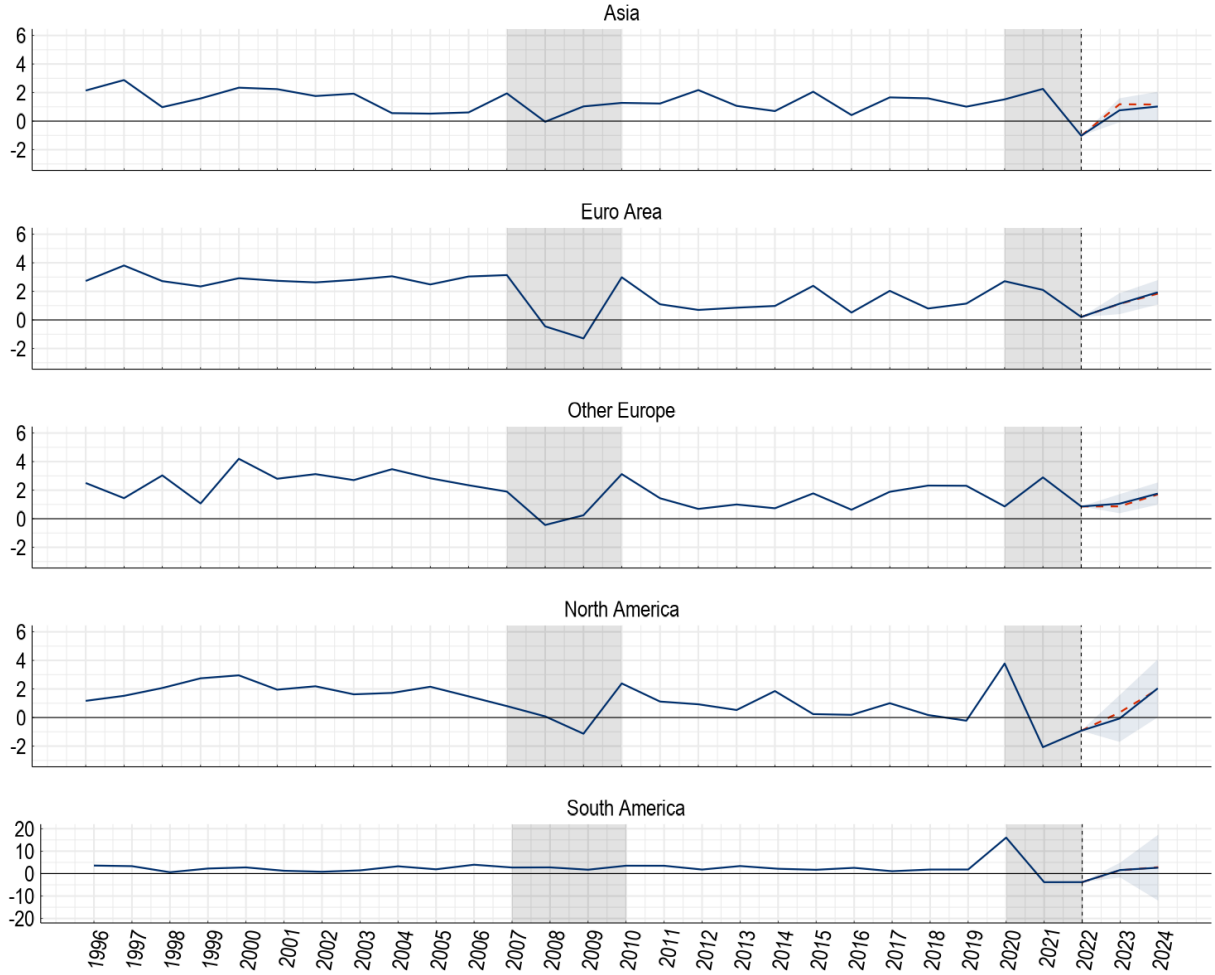
36. The rebound in labour productivity growth is estimated to be of various magnitudes in the different regions (Figure 9):

- The uptick in productivity growth in 2023 is estimated to have been marked in Asia, where productivity growth was negative or close to zero in 2022 in most countries covered in the analysis. Productivity growth would be flat thereafter.
- Labour productivity growth is also expected to recover in 2023 in North America but to slow rates, before rising to close to 2% the following year. Developments in South America appear to be particularly uncertain.
- On surface, productivity performance in Europe is found to be relatively good. But this masks a wide diversity within the region. Performance appears to be boosted by strong growth in small Nordic or Central Eastern European countries. In some Baltic countries, the dynamism of labour productivity growth in 2023-24 reflects to a large extent strong FDI inflows. Labour productivity growth is found to recover steadily in Germany and Italy but to improve only slowly in France.

37. Uncertainties around nowcasts are large, particularly in small open economies (Figure 10). Those metrics have been computed using model performance in the test set. As expected, confidence bands are larger for 2024 than for 2023. In particular, countries with the most volatile target variables are characterised higher by the larger uncertainties, as depicted by the magnitude of the confidence bands. Nowcasts in Ireland, Costa Rica and Colombia stand out as being particularly uncertain, reflecting the lack of accuracy of the best selected models in those countries.

Figure 9. Labour productivity growth in the OECD and by region

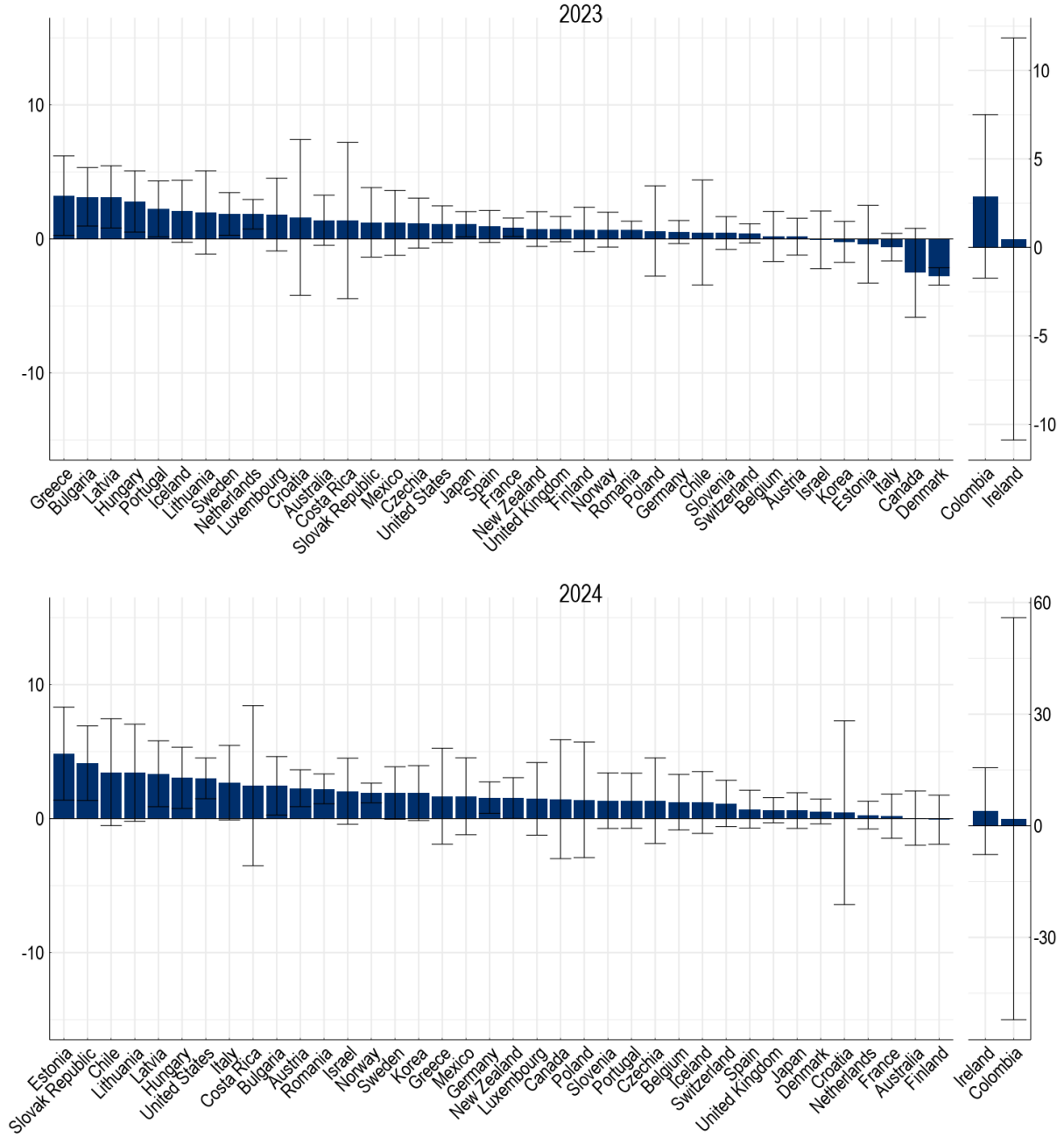
Region-specific unweighted average of labour productivity growth, percentage



Note: The grey area refers to the 95% confidence bands associated with each predicted point. They are computed as $\left[\hat{\mu} \pm p_{97.5}\, \hat{\sigma}/\sqrt{N}\right]$ where $\hat{\mu}$ corresponds to the unweighted average prediction and $\hat{\sigma}$ the average RMSE, $p_{97.5} \approx 1.95$ is the 97.5th percentile of the normal distribution, and N the number of countries in the respective region. Red dotted lines correspond to the consensus prediction. The red dotted line depicts the consensus nowcast. Asia: Korea, Japan, Australia and New Zealand. Euro area: Austria, Belgium, Croatia, Estonia, Finland, France, Germany, Greece, Italy, Ireland, Latvia, Lithuania, Luxembourg, the Netherlands, Portugal, Slovakia, Slovenia and Spain. Other Europe: Denmark, Bulgaria, Romania, Switzerland, Hungary, Iceland, Poland, Sweden and Norway. North America: United States, Canada and Mexico, South America: Chile, Colombia and Costa Rica. Predictions are obtained from selected best models and MIDAS methods (See section 2 for a complete description).
Source: Author's calculations.

Figure 10. Nowcast by country

Predicted labour productivity growth per country, percentage



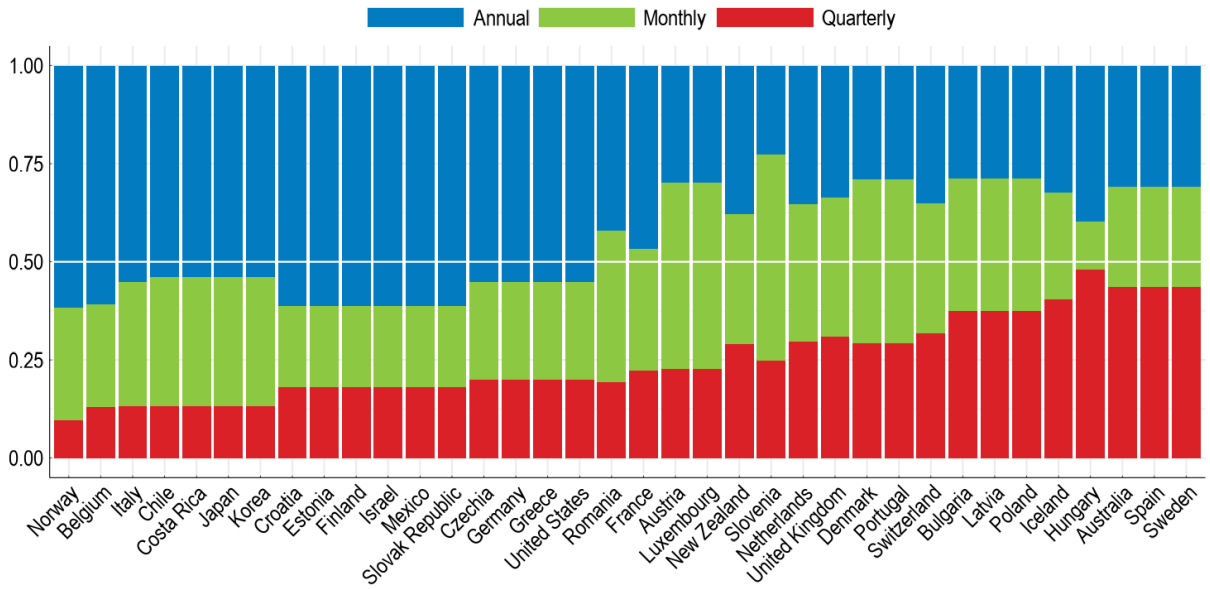
Note: Predictions are obtained from selected best models and MIDAS methods (See section 2 for a complete description). Details on estimates and performance metrics for the 1-step and 2-steps ahead exercises are available in Annex E and Annex F respectively. Vertical lines refer to the 90% confidence bands associated with each predicted point. They are computed as $[\hat{\mu} \pm p_{97.5}\hat{\sigma}]$ where $\hat{\mu}$ corresponds to the prediction and $\hat{\sigma}$ the RMSE computed over past predictions, $p_{97.5} \approx 1.95$ is the 97.5th percentile of the normal distribution.
Source: Author's calculations.

38. Relative importance helps to open up the black box and offers insights about the underlying factors driving the prediction outcomes. The relative importance of a feature is computed of as the average reduction in nowcast errors (measured in terms of RMSE) of GBM models allowed by the feature. Assessing feature importance, however, does not necessarily indicate a causal relationship between the dependent variables and the regressors, nor does it clarify the direction of the relationship.

39. In most countries, higher frequency data either at monthly or quarterly frequency are key determinants of the nowcast, explaining about 50-75% of the outcomes in most countries (Figure 11), which highlights the usefulness of the MIDAS framework. Both global and country-specific indicators are estimated to have been useful in nowcasting labour productivity growth in OECD countries (Figure 12). The relative importance of country-specific indicators varies across countries. Iceland stands out, with only country-specific features contributing to the nowcast. The share of country-specific variables is also higher than average in the United States, Greece, Germany, Czech Republic and Hungary.

Figure 11. Feature importance by frequency

Average feature importance by country and frequency

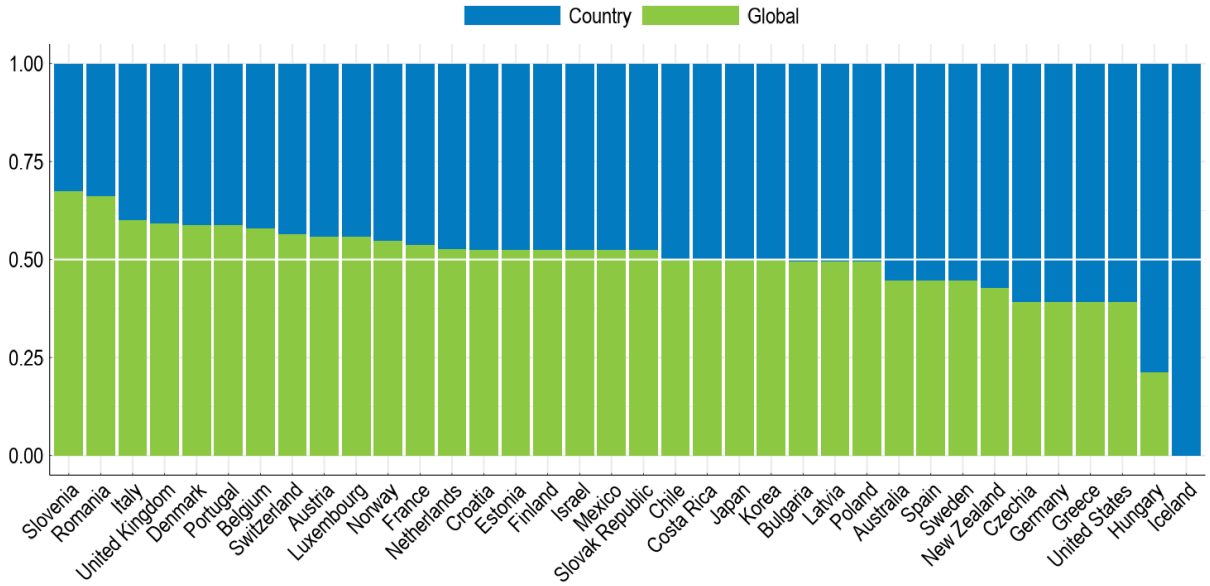


Note: The mean decreases in node RMSE are averaged over features of the same country and frequency. The resulting values are then normalised to sum to one at the country level. Feature importance values are obtained from the country-specific best model of the imposed-weights MIDAS method and calculated over the period 2015-22.
Source: Authors' calculations.

40. In most countries, activity and financial variables are found to be the indicators of most importance (Figure 13). By contrast, indicators related to the labour market situation, or the external environment played a lesser role. The lagged values of labour productivity growth emerge as the variable with the highest predictive power across countries, with the degree of importance varying across them (Figure 14). Its strongest influence in nowcasting labour productivity growth is observed in Japan and Portugal. Hours worked rank as the second most important factor, which is understandable given its essential role in the derivation of productivity. The third most influential group of variables includes expected financial volatility, real exchange rates, and gross fixed capital formation.

Figure 12. Feature importance by variable scope

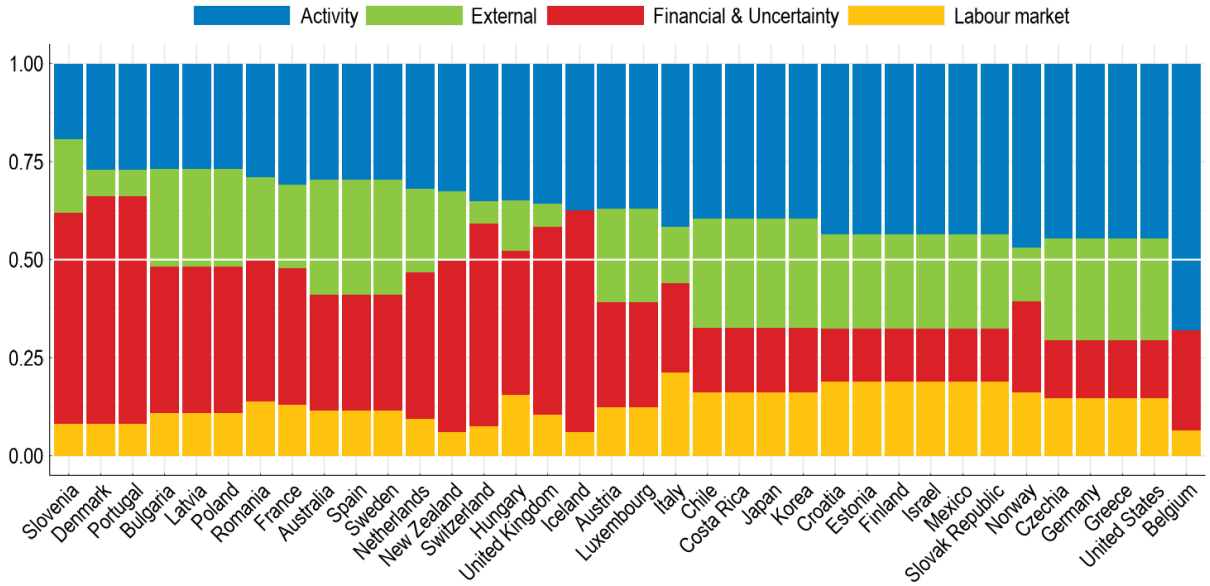
Average feature importance by country-specific and global features



Note: The mean decreases in node RMSE are averaged over features of the same country and scope. The resulting values are then normalised to sum to one at the country level. Feature importance values are obtained from the country-specific best model of the imposed-weights MIDAS method and calculated over the period 2015-22.
Source: Authors' calculations.

Figure 13. Feature importance by domains

Average feature importance by domains



Note: The mean decreases in node RMSE are averaged over features of the same country and sector. The resulting values are then normalised to sum to one at the country level. Feature importance values are obtained from the country-specific best model of the imposed-weights MIDAS method and calculated over the period 2015-22.
Source: Authors' calculations.

Figure 14. Country-specific feature importance

Mean decrease in RMSE in the GBM

Note: GBM operates by constructing multiple small regression trees and aggregating their predictions. Each single tree attempts to partition the input set, one node at a time, into groups of similar observations. Therefore, variable usefulness can be measured by the marginal improvement a variable offers in the partitioning of the input set. This marginal improvement is measured by the decrease in RMSE (Root Mean Squared Error). The mean decrease is then computed over all trees. Estimates are obtained from selected best models of the imposed-weights MIDAS method and calculated over the period 2015-22.
Source: Authors' calculations.

6. Conclusion

41. This paper has developed a methodology to nowcast labour productivity growth, combining three distinctive features: the use of a range of models, including several variants of machine learning approaches, mixed frequency models and panel settings. Models are selected according to their quasi-real time one-year ahead prediction performance.

42. Results are on average across countries very promising and provide useful insights of productivity developments in a very timely manner. The average error for the one-year-ahead prediction of labour productivity growth is less than 10% on average over the period 2015-22 and across OECD countries. Models using GBM appear to outperform alternatives and mixing frequency improves the nowcast accuracy.

43. Model performance varies across countries, though, with largest errors often found for small volatile economies. Model performance still remains unsatisfactory for Ireland, Colombia and to a lesser extent Croatia and Costa Rica. Country-specificity not captured in the current set of features may explain those under performance and enriching the feature dataset could be further investigated to improve model performance of those countries.

44. An examination of feature importance suggests that high-frequency variables contribute substantially to the nowcasts, especially those pertaining to economic activity and the external environment, including political stability and international trade disruptions have played. This also indicates the MIDAS framework can help in providing timely and more informed estimates.

45. The process was developed to allow for transparent and relatively easy annual updates. Although the pipeline was designed the nowcast labour productivity growth, it can be easily adapted to other economic variables such as trade or business statistics, which are often closely monitored to inform real time business cycle analysis.

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Annex A. Literature review

Nowcasting, which collapses in one term 'now' and 'forecasting', usually refers to the prediction of economic subjects in current time, the very near future and the very recent past. The basic principle of nowcasting is to exploit the information which is published early and possibly at higher frequencies than the variable of interest to obtain an 'early estimate' before the official figure becomes available (Bánbura et al., 2013). Nowcasting techniques are widely employed in applied macroeconomic research to predict aggregate GDP growth (Dauphin et al., 2022; Fosten and Greenaway-McGrevy, 2022), inflation (Modugno, 2012; Macias, Stelmasiak and Szafranel, 2022), or trade (Jaax, Gonzales and Mourougane, 2024; Dubovik et al., 2022; Chin, Meunier and Stumpner, 2023).

There have been scarce attempts to nowcast productivity

Indicators of productivity are widely used to inform policymaking and monitor the impact of reforms. Although those statistics are available with a delay of one or two years, depending on the countries, attempts to nowcast most recent developments in productivity growth have been scarce. A handful of recent papers have focused on productivity, very often looking at micro-level productivity of specific occupations or industries (e.g. Goodarzizad et al. (2023) for the construction sector, Tam et al. (2002) for excavation, Hanna et al. (1999) for electrical construction). However, to our knowledge, there has been no attempt to nowcast macro-economics measures of labour productivity growth in a large group of countries, which is the focus of this paper.

A variety of methods from the simplest to the most sophisticated ones have been employed to nowcast productivity. Zmuck et al. (2018) forecast labour productivity using a range of statistical methods: the status quo naïve model, the difference naïve model, the rate of change naïve model, geometric mean forecasting model, simple average forecasting model, single exponential smoothing model, and linear trend model. At the other hand of the spectrum, Mirahadi and Zayed (2016), Khan (2005), Wang (2005) and Nasirzadeh et al. (2020) used regression analysis, artificial neural networks (ANNs) or neuro-fuzzy system. ANNs are considered best suited to predict productivity considering their dynamic learning mechanisms with effective recognition capabilities (Yi and Chan, 2014).

Machine and deep learning offer a useful addition to standard nowcasting approaches

In recent years, the demand for more timely information to complement official statistics has regained importance. Amongst the most standard nowcasting models, Dynamic Factor Model (DFM) were first introduced in Geweke (1977) and Sargent et al. (1977) and have become very popular (Stock and Watson, 2006; Giannone et al., 2008; Banbura and Runstler, 2011). The underlying intuition behind those models is that the dynamics of a large set of time series can be explained by a small number of factors with an autoregressive dependance (Forni et al., 2005; Doz et al., 2011; Stock and Watson, 2016).

When cast in a state-space form and estimated using Kalman filtering, these models allow to handle the irregularities of the data in real time (i.e. mixed frequencies and non-synchronicity of the data releases). There are some limitations related to DFMs though, as estimating those models in a multi-frequency setting is found to be computationally heavy. In addition, the factor number selection in such settings has not been sufficiently addressed in the literature (Ballarin et al., 2023).

Easier access to alternative data together with enhanced computer power and storage capacity has facilitated the recourse to machine and deep learning in nowcasting. The performance of machine learning models in accurately nowcasting macro-economics indicators has been examined in depth. This is often done by testing models on different types of datasets and examining the out-sample properties of a range of nowcasting models, usually specific machine learning approaches and DFMs (e.g. Smalter Hall, 2018). Most of the literature has focused on Anglo-saxon economies, but the coverage has recently expanded to Europe (e.g. Gogas et al. 2022) and Japan (Maehashi and Shintani, 2020).

Machine-learning techniques appear to often win the horse race, though not systematically. Within the 'heterogeneous' group of machine learning techniques, gradient boosting has demonstrated some promising properties. Evidence tends to suggest that no approach would systemically outperform the other, with performance depending inter alia on the sparsity of the dataset (Giannone, Lenza and Primiceri, 2021). Following this logic, Goulet Coulombe et al. (2022) and Vedder and van de Winkel (2024) seek to understand how features such as non-linearity, data richness, cross-validation and shrinkage help to explain the predictive power of the best-performing models. The former paper found evidence for US data that non-linearity is often key to explain good machine-learning nowcasting performance. The second refines those conclusions and show that, for the Netherlands, in environment with limited data, the optimal combination of features is specific to the variable that is predicted.

Recent developments include the use of mixed-frequency models

The increasing availability of data, of different frequency, has naturally encouraged researchers to combine this information in an efficient way to improve the accuracy of nowcasting models. The simpler approach is to convert high-frequency data, without any adjustment, into lower frequency, for instance averaging quarterly series into annuals. While simple, this results in ignoring key information (e.g. ignoring the first quarters of the years when dealing with annual data and the last quarter is missing).

To incorporate most recent data points, early approaches relied on "bridge models" i.e. using linear regressions that link high frequency to low frequency variables. These models are used to forecast the high-frequency indicators, which is then converted to a lower frequency and used as an input in the low-frequency-based model (Baffigi, Golinelli and Parigi, 2004). One limitation of these approaches is that they lose some of the information contained in the high-frequency series, when the latter is aggregated (i.e. when a specific pattern was observed in the course of the year). Another limitation is that, depending on the target variables and the features, the information content of the bridge may sometimes be limited. Alternative such as state-space models, estimated using the Kalman filter, where the low-frequency variable is considered as a high-frequency one with missing information, are less common.

Mixed-data sampling (MIDAS) regression models developed by Ghysels et al. (2004) makes a direct use of mixed frequency data and account for all the information contained in high-frequency series to predict lower frequency variables. MIDAS is horizon dependent and needs to be re-estimated for each forecast horizon. The standard MIDAS approach imposes a lag-polynomial structure on the high-frequency data to reduce the number of parameters that need to be estimated. By contrast the unrestricted version (U- IDAS) allows each high frequency feature to have its own coefficient: this makes interpretation easier but at the cost of estimating more coefficients (Foroni, Marcellino and Schumacher, 2013). See Foroni and Marcellino (2013) for a survey of econometric methods for mixed-frequency data.

While there is consensus that exploiting data at different frequency matters, it is not clear which method is superior (Foroni and Marcellino, 2013). States-space approaches generally work but only if the number of variables in the model is small. An increase in the number of parameters rises dramatically the complexity of the model. By contrast, MIDAS models appear to be more robust to misspecification compared both to bridge equations models and state-space approaches, and computationally simpler.

Attempts to combine mixed frequency models and machine learning have been limited

Despite a clear trend toward the use of machine and deep learning on the one hand, and to handle model with mixed frequency on the other hand, there has been only a few examples combining the two approaches. One such example is MLMF, which merges U-MIDAS with machine learning, for both linear (LASSO, elasticnet) or non-linear models (Artificial neural networks). Borup, Rapach and Schütte (2023) illustrate the value of the MLMF approach by generating a sequence of nowcast and backcasts of US weekly unemployment insurance initial claims based on google trends search data. In the same vein, Jaax et al. (forthcoming) use machine learning and Midas to nowcast services trade with a focus on some industries. Another example can be found in Kamolthip (2021) who combines MIDAS with a Long-Short-Term Memory network (LSTM).

Annex B. Setting the data in a MIDAS framework

This MIDAS was initially developed by Ghysels, Santa-Clara, and Valkanovand (2004). It allows the use of explanatory variables of various frequencies. The original approach was adapted to be combined with a panel setting following Borup et al. (2023).

The objective is to nowcast $y_{c,t}$ the target variable (labour productivity growth) for a country c at time t. using its past values and a set of explanatory variables $X_t^{(m)}$. X combines variables of different frequencies $m \in M$ where $M = \{1, 4, 12\}$ is the set of frequencies included in the model. θ the parameters of the MIDAS polynomials (see below).

$$y_{c,t} = f\left(y_{c,t-1}, \left\{X_{c,t}^{(m)}\right\}_{m \in M}; \theta\right) + \varepsilon_{c,t} \tag{1}$$

Or

$$y_{c,t} = f\left(y_{c,t-1}, X_{c,t}^{(1)}, X_{c,t}^{(4)}, X_{c,t}^{(12)}; \theta\right) + \varepsilon_{c,t} \tag{2}$$

Where $X_{c,t}^{(1)}$, (resp. $X_{c,t}^{(4)}, and\ X_{c,t}^{(12)}$) represents the set of variables sampled at the annual (resp. quarterly, and monthly) frequency. In other words

$$X_{c,t}^{(m)} = \left[x_{1_{c,t}}^{(m)\prime} \quad x_{2_{c,t}}^{(m)\prime} \quad \cdots \quad x_{K_{m_{c,t}}}^{(m)}\right]' \tag{3}$$

Is a $K_m \times T$ matrix, where K_m is the number of predictors of frequency m, and

$$x_{i_t}^{(m)} = \left[x_{i_{t+1/m}}^{(m)} \quad \cdots \quad x_{i_{t+j_m/m}}^{(m)}\right] \tag{4}$$

A $m \times 1$ vector whose elements correspond to the lags of the ith variable of frequency m.

The introduction of high frequency variables in the model results in an increase in the number of parameters. For example, if $y_{c,t}$ is monthly and is explained by a single, daily regressor (i.e. m = 22), including six lags in the model would yield 132 parameters, under the most parsimonious specification ($K_m = 1$ and $|M| = 1$).

This situation, often referred to as the curse of dimensionality, complicates the model estimation. The core idea of MIDAS is to circumvent this by imposing a lag structure on higher frequency variables using lag polynomials. These polynomials are functions of a small number of parameters. Using the conventional notation: $L^k x_t := x_{t-k}$ and $A(L)x_t := \sum_{i=1}^{i_{max}} a_i x_{t-i}$,

$$B\left(L^{-\frac{1}{m}}; \theta_B\right) x_t^{(m)} = b(1, \theta_B) x_{t+1/m}^{(m)} + b(2, \theta_B) x_{t+2/m}^{(m)} + \cdots + b(j_m, \theta_B) x_{t+j_m/m}^{(m)} \tag{5}$$

For identification purposes, coefficients are usually normalised such that $\sum_{i=1}^{L} b_i = 1$. B can be chosen to accommodate the nature of a particular variable, for example, setting $B(L) = \mathbf{1}\{i = i^*\}$ corresponds to point-in-time sampling, an appropriate structure for stock variables (Foroni et al, 2018).

The conventional nowcasting practice which consists in aggregating the high-frequency variable by taking the average can be seen as a particular form of the restricted MIDAS. Specifically, this corresponds to $B(L^{-1}) = \frac{1}{j_m}\sum_{j=1}^{j_m} x_{t+j}$, and is referred as imposed weights MIDAS in the paper.

Likewise, the unrestricted version of the MIDAS, U-MIDAS, corresponds to an estimation where $B(L) = I$, the identity function.

Lag polynomials can also be chosen to flexibly fit the target in a data-driven way. The Beta polynomial (Ghysels et al., 2007) is a typical example:

$$b(k;\theta_1,\theta_2) = \frac{f(\frac{k}{K};\theta_1,\theta_2)}{\sum_{k=1}^{K} f(\frac{k}{K};\theta_1,\theta_2)} \tag{6}$$

Where $f(x;\theta_1,\theta_2) = \frac{x^{\theta_1-1}(1-x)^{\theta_2-1}\Gamma(\theta_1+\theta_2)}{\Gamma(\theta_1)\Gamma(\theta_2)}$, and $\Gamma(\theta_1) = \int_0^\infty e^{-x}x^{\theta_1-1}dx$.

In this context, equation (2) becomes:

$$y_{c,t} = f\left(y_{c,t-1}, X_{c,t}^{(1)}, B_Q\left(L^{\frac{1}{4}};\theta_Q\right)X_{c,t}^{(4)}, B_M\left(L^{\frac{1}{12}};\theta_M\right)X_{c,t}^{(12)};\delta\right) + \varepsilon_{c,t} \tag{7}$$

The estimation is performed following the Profile Likelihood method suggested in Ghysels and Qian (2018). The estimation of MIDAS can be difficult, especially in the context of machine learning. Indeed, the hyperparameters of the model, δ, and the parameters of the MIDAS polynomials, θ, must be estimated jointly, which can be computationally demanding. The profile likelihood estimation aims to alleviate this difficulty by first fixing $\theta = \bar{\theta}$, and then optimising $\hat{\delta} = \delta(\bar{\theta})$. Finally, θ is obtained as $\hat{\theta} = \theta(\hat{\delta})$. While the problem, as stated in Ghysels and Qian (2018), boils down to a closed-form solution and is estimated with OLS, the complexity of our models requires a resolution by numerical optimisation.

Annex C. Description of the models

Several nowcasting approaches were tested and compared to an autoregressive model (AR) of order one, that is used as a benchmark. This following section provides a brief description of each model covered in this study.

Regularised Regressions (Ridge, Lasso, and Elastic Net)

Regularised regressions extend linear models by adding a penalty term to the objective function in order to control model complexity and improve predictive stability (Tibshirani, 1996; Hoerl and Kennard, 1970), which is particularly important under the MIDAS framework where the number of explanatory variables X_t can be large. We assume:

$$y_t = X_t\beta + \varepsilon_t$$

Where y_t is productivity growth at time t, X_t is the design matrix of features, β the vector of coefficients, and ε_t is an error term. The OLS approach seeks to minimise the sum of squared residuals, $\sum_{t_0}^{T}(y_t - X_t\beta)^2$, regularised regression techniques, on the other hand, incorporate an additional penalty term into this minimisation problem:

$$\min\left\{\sum_{t_0}^{T}(y_t - X_t\beta)^2 + \lambda P(\beta); \beta\right\}$$

Where $\lambda \geq 0$ is a parameter regulating the strength of the penalty $P(\beta)$. Different choices of $P(\beta)$ give rise to different forms of regression. In ridge, often referred to as L_2 regularisation, the penalty is the sum of squared coefficients, $P_{Ridge}(\beta) = \sum_j \beta_j^2$. This shrinks all coefficients toward zero and is particularly effective in handling highly correlated predictors, although it does not drive any coefficients exactly to zero. By contrast, lasso regression uses an L_1 penalty, $P_{Lasso}(\beta) = \sum_j |\beta_j|$, which can drive some coefficients exactly to zero. This property effectively causes variable selection, an often-desirable mechanism in nowcasting with many features since it helps isolate the most relevant predictors. Elastic net combines the L_1 and L_2 penalties through a mixing parameter α, so that:

$$P_{EN}(\beta) = \alpha\sum_j |\beta_j| + (1-\alpha)\sum_j \beta_j^2$$

This blend allows elastic net to perform variable selection while also retaining the stabilising influence that ridge provides when predictors exhibit high collinearity.

In practice, the penalty term λ (and the mixing parameter α for elastic net) is typically chosen via cross-validation. Such tuning ensures that the model neither overfits idiosyncratic short-term fluctuations nor overlooks essential explanatory variables.

Dynamic Factor Model (DFM)

The Dynamic Factor Model (Stock and Watson, 2010) is built on the premise that a potentially large set of observed explanatory variables can be well-described by a smaller number of latent common factors. This

compression is especially valuable in macroeconomic environments, where variables can be noisy and interconnected. By extracting a few underlying factors, the DFM captures the core co-movements across features and uses them to explain and predict the target, that is:

$$y_t = F_t\beta + \varepsilon_t$$

Where F_t denote a vector of r latent factors, such that $r \ll \dim(X_t)$.

Those models have been found to be useful for nowcasting major economic aggregates (Doz and Fuleky, 2019) and have been widely used to predict global trade (Guichard and Rusticelli, 2011; Martínez-Martín and Rusticelli, 2020; Cantú, 2018).

The measurement equation of the DFM can be written as

$$X_t = \Lambda F_t + \xi_t$$

where Λ is a loading matrix that links the factors to the observed variables and ξ_t is an idiosyncratic component with specific characteristics for each variable (e.g. serial correlation or measurement error). The dynamics of F_t are described by a vector autoregressive (VAR) process:

$$F_t = \Psi F_{t-1} + \eta_t$$

Where Ψ is a coefficient matrix governing factor persistence and η_t is a white noise process of innovations. In this framework, the measurement equation corresponds to how X_t depends on F_t, while the transition equation encapsulates the dynamics of the latent factors. All parameters are estimated with a Kalman filter through a cross-validation exercise.

Because the DFM summarises the common patterns driving a large set of macroeconomic indicators, it can produce coherent predictions with fewer parameters than a full system of individual equations, this is particularly relevant in a setting like MIDAS where the dimensionality of the input set grows rapidly.

Tree-Based Models (Random Forest and Gradient Boosting)

Tree-based models are nonparametric methods that recursively partition the input space to make predictions for y_t. They offer a flexible way to capture potentially complex, nonlinear relationships between X_t and y_t.

A regression tree is constructed by successively splitting the data into subgroups based on the values of the explanatory variables. At each split, the algorithm selects the variable and threshold that best reduce a chosen measure of error (e.g. mean squared error) within the resulting partitions. This process continues until some stopping criterion is met, producing a tree whose leaves correspond to distinct regions of the input space, each associated with a predicted value for y_t. Because of its hierarchical structure, a single regression tree can capture interaction effects among explanatory variables without explicitly specifying them, although it may suffer from high variance and a tendency to overfit.

One way to address these limitations is to combine multiple trees through an ensemble strategy known as bagging (Bootstrap Aggregation). Bagging builds many trees on bootstrapped samples of the training data, then aggregates their predictions (e.g. by averaging) to obtain a more robust final estimate. We cover two widely used approaches to constructing ensembles of regression trees—Random Forest and Gradient Boosting.

Random Forest

Random Forest (Breiman, 2001) extends the idea of bagging by injecting additional randomness in the tree-building process. Instead of evaluating splits on every available explanatory variable, each tree is restricted to a random subset of variables at each potential split. This strategy reduces the correlation

among individual trees, making the aggregated predictions more stable. This process can be seen as the estimation of an underlying "true" tree (Goulet-Coulombe, 2024).

Formally, for each of the B trees in the forest, a bootstrap sample of the training data is drawn and a random subset of the explanatory variables is considered at each split. The resulting regression trees, each of which might emphasise different aspects of the data, are then averaged to produce the final prediction for y_t.

$$y_t = \frac{1}{B}\sum_{b=1}^{B} T_b(X_\mathrm{t})$$

Where $T_b(X_\mathrm{t})$ corresponds to the estimate of the bth regression tree. This mechanism enhances the model's ability to generalise, as it diminishes the risk that a small group of highly predictive variables will dominate all splits. Random Forests are often favoured when one needs robust predictions that can nonlinearly handle a large number of potentially correlated and partially redundant indicators.

Gradient Boosting

Gradient Boosting (Friedman, 2001) also involves an ensemble of regression trees but constructs them in a sequential manner, where each new tree is designed to improve the performance of the ensemble by targeting the current residuals.

Let $\hat{y}_t^{(m)}$ be the prediction of the ensemble after m trees. A new tree $\hat{T}^{(m+1)}$ is fitted to the residuals $\hat{r}_t^{(m)} = y_t - \hat{y}_t^{(m)}$. The overall prediction is then updated as follows:

$$\hat{y}_t^{(m+1)} = \hat{y}_t^{(m)} + \nu\hat{T}^{(m+1)}(X_\mathrm{t})$$

where the learning rate, ν, controls how rapidly each new tree corrects the errors of the current ensemble. By fitting each successive tree to the gradient of the loss function, the boosting algorithm concentrates on aspects of the prediction that remain challenging, leading to a refined model.

Both models can be customised by adjusting the way single trees are built. This is achieved by the estimation of hyperparameters, implemented through a cross-validation exercise. The maximum depth of trees limits how many sequential splits can occur. A deeper tree can capture more complex relationships X_t but it risks overfitting if it becomes too specific to particular instances. By contrast, shallower trees may overlook subtler patterns but generally exhibit lower variance. The minimum number of observations in each leaf sets a lower bound on how many samples must end up in any terminal node. Higher values encourage broader, more generalised leaves and can reduce overfitting, while very small leaves can make the trees highly sensitive to minor fluctuations in the data. Similarly, the minimum number of observations required to attempt a split dictates how many samples must be present in a node for a further split to be considered. Increasing this threshold prevents the model from pursuing narrowly focused splits on small subsets, effectively adding a layer of regularisation. Lastly, the maximum number of features considered at each split controls how many predictors may be evaluated in the partitioning process. Restricting this pool in random forest promotes diversity among individual trees, as each one relies on a different subset of variables. When more features are allowed at each split, the trees can become more specialised, but at the risk of overfitting to particular patterns in the data.

Our estimation process covers the entire space of hyperparameters and includes 1 200 candidate tree-based models. This is done to allow the comparison of different degrees and variants of flexible models as well as more robust ones.

Annex D. Information on the data

Table A D.1. Data sources

Category	Variable	Source	Frequency	Dimension
Target variable	GPD per hour worked in USD, constant prices, 2015 PPPs	Productivity Levels, OECD	A	Country
Financial development and uncertainty	Volatility index	cboe.com	M	Global
	World Uncertainty Index	policyuncertainty.com	Q	Global
	World Trade Uncertainty Index	policyuncertainty.com	Q	Global
	Global Supply Chain Pressure Index	NYFED	M	Global
	Worldwide governance indicators	DataBank, World Bank	A	Country
	Geo-political Risk Index	matteoiacoviello.com	M	Country
	Economic policy uncertainty index	Policyuncertainty.com	Q	Country
	M1 (currency and coins, plus overnight deposits)	Monetary aggregates, OECD	M	Country
	M3 (sum of M1, deposits with an agreed maturity of up to two years and deposits redeemable at notice of up to three months, repurchase agreements, money market fund shares and debt securities up to two years)	Monetary aggregates, OECD	M	Country
External environment	Total value of Imports of goods	Main Economic Indicators, OECD	A	Country
	Total value of Exports of goods	Main Economic Indicators, OECD	A	Country
	Openness to trade	Main Economic Indicators, OECD	A	Country
	Share of exports in the global total export	International Merchandise Trade Statistics, OECD	Q	Country
	Foreign Direct Investment, net	DataBank, World Bank	A	Country
Activity	Rate of capacity utilisation in manufacturing	Main Economic Indicators, OECD	Q	Country
	Production in Construction, index	Main Economic Indicators, OECD	Q	Country
	Number of permits issued for residential building, index	Main Economic Indicators, OECD	Q	Country
	Total production in manufacturing, index	Main Economic Indicators, OECD	Q	Country
	Total retail trade, index	Main Economic Indicators, OECD	Q	Country
	GDP in USD, current prices	Annual National Accounts, OECD	Q	Country
	Gross domestic product, market prices, deflator	National Accounts, OECD	Q	Country
	General government gross financial liabilities as a percentage of GDP	National accounts, OECD	A	Country
	Government final consumption expenditure	Quarterly National Account, OECD	Q	Country

Category	Variable	Source	Frequency	Dimension
	Gross fixed capital formation	Quarterly National Account, OECD	Q	Country
	Share of ICT goods as percentage of trade	UNCTAD Digital Economy Database	A	Country
	Government budget allocations for R&D	Government budget allocations for R&D, OECD	A	Country
	Total gross debt as percentage of GDP	Quarterly public sector debt, OECD	Q	Country
	GDP per person employed	Unit Labour Cost, OECD	Q	Country
	Average annual hours worked per worker	Average annual hours actually worked per worker, OECD	A	Country
	Fixed broadband in DSL subscriptions per 100 inhabitants	Broadband and telecom databases, OECD	A	Country
	Fixed broadband in all subscriptions per 100 inhabitants	Broadband and telecom databases, OECD	A	Country
	Number of patents	Google BQ	M	Country
Labour market and prices	Compensation of employees	National Accounts, OECD	A	Country
	Participation in labour market: Unemployment rate, Aged 15 and over, All persons	Intra-annual labour statistics, OECD	Q	Country
	Participation in labour market: Activity rate, Aged 15 and over, All persons	Intra-annual labour statistics, OECD	Q	Country
	Consumer Price Index (All items)	Main Economic Indicators, OECD	Q	Country
	Real effective exchange rate, constant trade weights	Main Economic Indicators, OECD	Q	Country
	Unemployment rate: all persons, s.a.	Key Economic Indicators, OECD	M	Country
	Employment, all persons	Annual employment by economic activity, domestic concept, OECD	Q	Country
	Duration of unemployment < 1 month	Unemployment by duration, OECD	A	Country
	Duration of unemployment > 1 month and < 3 months	Unemployment by duration, OECD	A	Country
	Duration of unemployment > 3 month and < 6 months	Unemployment by duration, OECD	A	Country
	Duration of unemployment > 6 month and < 1 year	Unemployment by duration, OECD	A	Country
	Duration of unemployment 1 year and over	Unemployment by duration, OECD	A	Country
	Number of enterprises in business economy, except financial and insurance activities	Structure business statistics by size class and economic activity, OECD	A	Country
	Turnover in business economy, except financial and insurance activities	Structure business statistics by size class and economic activity, OECD	Q	Country

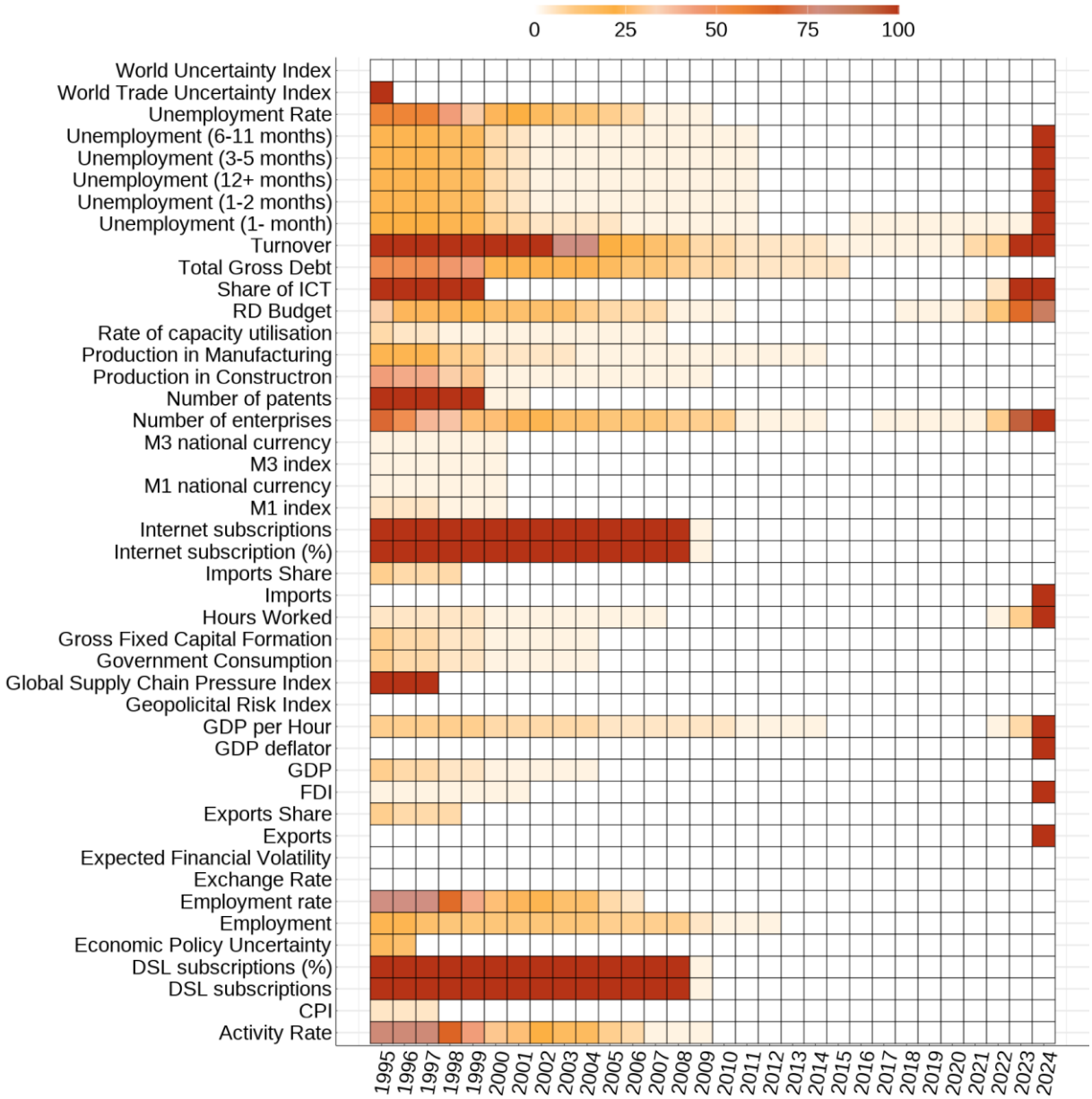
Figure A D.1. Missing structure per country

Percentage of missing values across countries and features

Note: The darker a cell, the more missing values there are for the corresponding country and feature.
Source: Authors' calculations.

Figure A D.2. Missing structure per year

Percentage of missing values across features over time



Note: The darker a cell, the more missing values there are for the corresponding country and year.
Source: Authors' calculations.

Figure A D.3. Overview of the global economy in 2023

Growth rate of the input set in 2023, percentage

Note: Heatmap of the observed growth rates of the variables included in the input set in 2023. A blue (red) cell corresponds to an increasing (decreasing) variable, a white cell corresponds to a stagnating variable. The strength of the variation is implied by the opacity of the corresponding cell.

Source: Authors' calculations.

Different imputation methods are applied based on the type of data missingness. When a variable has partial missing data for a given country, the median value of the series for that country is used to fill in the gaps. In case of complete missingness, the country-level series is imputed using information from a neighbouring country. If data of neighbouring country is not available, imputation is performed using data from the geographical region to which the country belongs.

For global variables, a uniform value is applied across all countries for each year. For country-level variables, if a Nordic country experiences a complete missing for a series, the average from other Nordic countries is used for imputation. If data is missing for a European country, it is imputed using the average value from the European region. For Türkiye, the average value of the OECD is applied. If an Asian country has entirely missing data, it is filled in with the average value of Asia region (with information for Japan replaced by Korea and vice versa). Similarly, missing data for Australia is imputed using data from New Zealand, and vice versa.

For missingness resulted from publication lags, the missing values are estimated using the nowcast growth of the variable. The nowcasting process to obtain this growth follows the same pre-processing and procedure outlined in section 2. Each feature is treated as the new target in the nowcast exercise. For each feature, the process involves data pre-processing, model training and testing to identify the best performance model across countries and calculating the growth rate based on the best model.

Annex E. Additional results for one-step ahead predictions

Table A E.1. Performance relative to the AR model

	Model	Method	RMSE	RMSE (excl. 2020-21)	RRMSE		RRMSE (excl. 2020-21)	FDA
Australia	GBM	Restricted	0.011	0.012	0.527	***	0.494	87.50%
Austria	GBM	Imposed Weights	0.008	0.009	0.704	***	0.902	62.50%
Belgium	GBM	Unrestricted	0.011	0.011	0.843	***	1.075	75%
Bulgaria	GBM	Restricted	0.013	0.013	0.723	***	0.789	100%
Canada	Ridge	Unrestricted	0.02	0.008	0.697	**	0.512	87.50%
Switzerland	GBM	Restricted	0.004	0.005	0.419	***	0.448	100%
Chile	GBM	Unrestricted	0.024	0.019	0.449	***	0.476	87.50%
Colombia	AR	Restricted	0.028	0.022	1	***	1	87.50%
Costa Rica	GBM	Unrestricted	0.035	0.023	0.652	***	0.758	100%
Czechia	GBM	Imposed Weights	0.011	0.012	0.501	***	0.491	100%
Germany	GBM	Restricted	0.005	0.004	0.939		0.625	100%
Denmark	GBM	Restricted	0.004	0.003	0.324	***	0.238	100%
Spain	DFM	Imposed Weights	0.007	0.006	0.846	***	0.662	62.50%
Estonia	GBM	Restricted	0.018	0.019	0.497	***	0.548	87.50%
Finland	GBM	Unrestricted	0.01	0.011	0.696		0.668	75%
France	GBM	Imposed Weights	0.004	0.005	0.364	***	0.472	100%
United Kingdom	GBM	Unrestricted	0.006	0.006	0.472	***	0.591	87.50%
Greece	GBM	Unrestricted	0.018	0.02	0.552	***	0.537	100%
Croatia	GBM	Imposed Weights	0.035	0.019	0.757	**	0.917	75%
Hungary	GBM	Unrestricted	0.014	0.016	0.674	**	0.749	87.50%
Ireland	Ridge	Unrestricted	0.069	0.065	0.886		0.914	87.50%
Iceland	GBM	Unrestricted	0.014	0.008	0.848		0.663	87.50%
Israel	GBM	Unrestricted	0.013	0.007	0.564	***	0.491	100%
Italy	GBM	Restricted	0.006	0.003	0.535	***	0.557	100%
Japan	GBM	Unrestricted	0.006	0.006	0.567	***	0.808	87.50%
Korea	GBM	Unrestricted	0.009	0.01	0.828	***	0.833	87.50%
Lithuania	Ridge	Unrestricted	0.019	0.021	0.505	***	0.491	75%
Luxembourg	GBM	Restricted	0.017	0.019	0.978	***	1.18	50%
Latvia	GBM	Restricted	0.014	0.01	0.673	***	0.58	100%
Mexico	GBM	Unrestricted	0.015	0.013	0.783	***	0.741	62.50%
Netherlands	GBM	Imposed Weights	0.007	0.006	0.739	***	0.735	62.50%
Norway	GBM	Restricted	0.008	0.005	0.607	***	0.358	87.50%
New Zealand	GBM	Imposed Weights	0.008	0.009	0.634	***	0.739	75%
Poland	GBM	Restricted	0.02	0.018	0.926	**	1.339	87.50%
Portugal	GBM	Unrestricted	0.013	0.014	0.695	***	0.672	50%
Romania	GBM	Unrestricted	0.004	0.004	0.41	***	0.456	100%
Slovak Republic	GBM	Unrestricted	0.016	0.014	0.598	***	0.482	87.50%
Slovenia	GBM	Imposed Weights	0.007	0.008	0.414	***	0.427	100%

	Model	Method	RMSE	RMSE (excl. 2020-21)	RRMSE		RRMSE (excl. 2020-21)	FDA
Sweden	GBM	Imposed Weights	0.01	0.011	0.542	***	0.569	75%
United States	GBM	Unrestricted	0.008	0.007	0.51	***	0.565	75%

Note: Models include the best performing model according to the RMSE (Root Mean Squared Error) criterion. RRMSE: Relative Root Mean Squared Error (obtained as the ratio of each model RMSE with that of the AR one) FDA: Forecast Directional Accuracy. Method corresponds to the MIDAS specification employed (See Annex B for more details). GBM: Gradient Boosting Trees (See Annex C for more details). Stars indicate the significance of the Diebold-Mariano (small sample) p-values as follows 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1. Tests include the Harvey-Leybourne-Newbold correction for small samples.

Table A E.2. Performance relative to the Dynamic factor Model

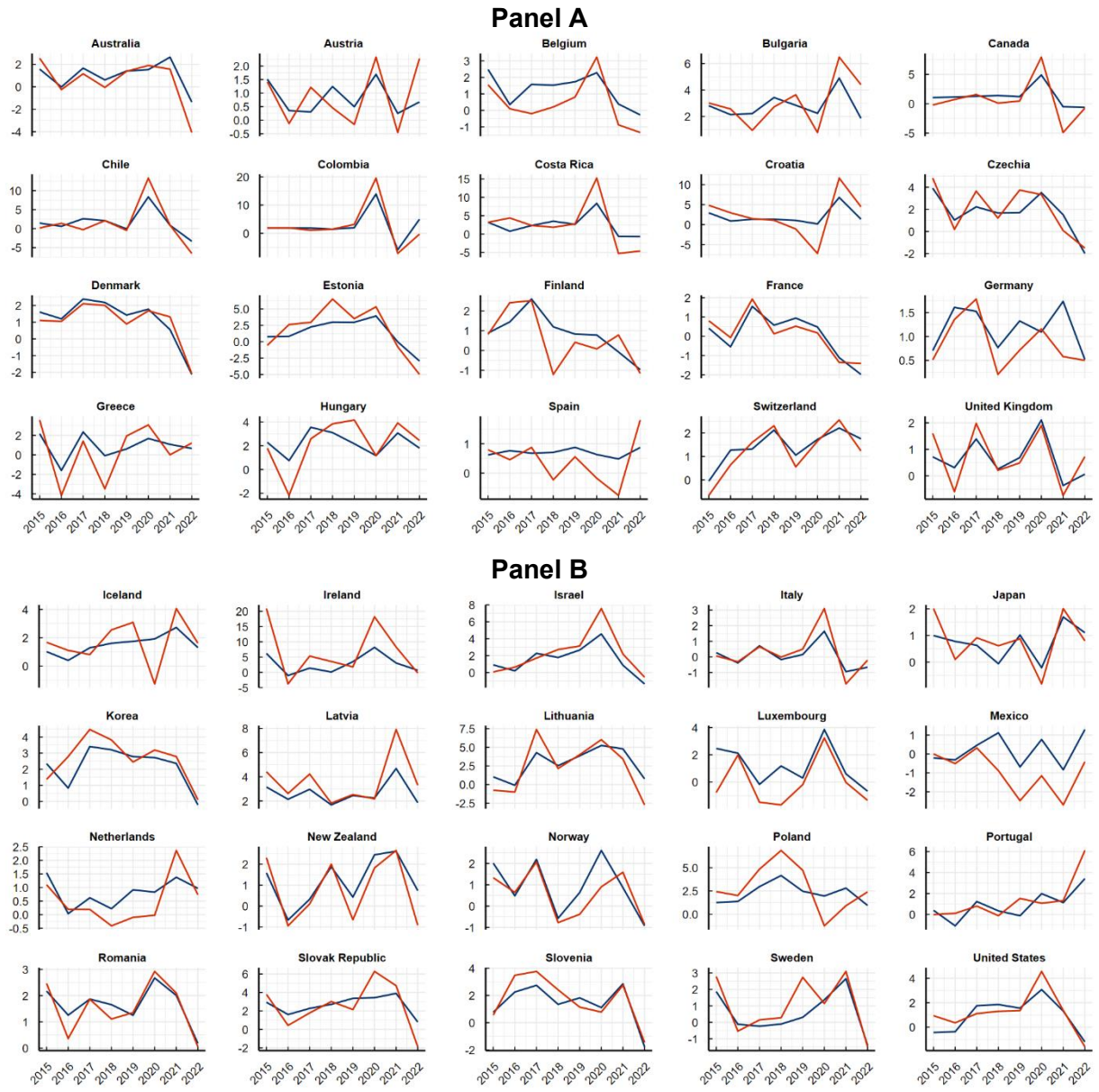
	Model	Method	RMSE	RMSE (excl. 2020-21)	RRMSE		RRMSE (excl. 2020-21)	FDA
Australia	GBM	Restricted	0.011	0.012	0.597	***	0.609	87.50%
Austria	GBM	Imposed Weights	0.008	0.009	0.797	***	0.91	62.50%
Belgium	GBM	Unrestricted	0.011	0.011	0.811	***	1.174	75%
Bulgaria	GBM	Restricted	0.013	0.013	0.735	***	0.86	100%
Canada	Ridge	Unrestricted	0.02	0.008	0.608	***	1.185	87.50%
Switzerland	GBM	Restricted	0.004	0.005	0.365	***	0.462	100%
Chile	GBM	Unrestricted	0.024	0.019	0.455	***	0.483	87.50%
Colombia	AR	Benchmark	0.028	0.022	0.999		0.997	87.50%
Costa Rica	GBM	Unrestricted	0.035	0.023	0.597	**	0.81	100%
Czechia	GBM	Imposed Weights	0.011	0.012	0.466	***	0.438	100%
Germany	GBM	Restricted	0.005	0.004	0.86		0.548	100%
Denmark	GBM	Restricted	0.004	0.003	0.279	***	0.224	100%
Spain	DFM	Imposed Weights	0.007	0.006	1		1	62.50%
Estonia	GBM	Restricted	0.018	0.019	0.483	***	0.522	87.50%
Finland	GBM	Unrestricted	0.01	0.011	0.369	***	0.394	75%
France	GBM	Imposed Weights	0.004	0.005	0.33	***	0.387	100%
United Kingdom	GBM	Unrestricted	0.006	0.006	0.584	***	0.708	87.50%
Greece	GBM	Unrestricted	0.018	0.02	0.676	***	0.677	100%
Croatia	GBM	Imposed Weights	0.035	0.019	0.667	**	0.763	75%
Hungary	GBM	Unrestricted	0.014	0.016	0.671	***	0.772	87.50%
Ireland	Ridge	Unrestricted	0.069	0.065	0.74	*	0.806	87.50%
Iceland	GBM	Unrestricted	0.014	0.008	0.42		0.267	87.50%
Israel	GBM	Unrestricted	0.013	0.007	0.525	***	0.494	100%
Italy	GBM	Restricted	0.006	0.003	0.511	***	0.387	100%
Japan	GBM	Unrestricted	0.006	0.006	0.523	***	0.656	87.50%
Korea	GBM	Unrestricted	0.009	0.01	0.391		0.447	87.50%
Lithuania	Ridge	Unrestricted	0.019	0.021	0.515	***	0.596	75%
Luxembourg	GBM	Restricted	0.017	0.019	0.407	**	0.421	50%
Latvia	GBM	Restricted	0.014	0.01	0.365	**	0.318	100%
Mexico	GBM	Unrestricted	0.015	0.013	0.326	***	0.312	62.50%
Netherlands	GBM	Imposed Weights	0.007	0.006	0.743	***	0.828	62.50%
Norway	GBM	Restricted	0.008	0.005	0.715	***	0.468	87.50%
New Zealand	GBM	Imposed Weights	0.008	0.009	0.562	***	0.711	75%
Poland	GBM	Restricted	0.02	0.018	0.579	***	0.449	87.50%
Portugal	GBM	Unrestricted	0.013	0.014	0.634	***	0.677	50%
Romania	GBM	Unrestricted	0.004	0.004	0.285	***	0.396	100%
Slovak Republic	GBM	Unrestricted	0.016	0.014	0.534	***	0.649	87.50%
Slovenia	GBM	Imposed Weights	0.007	0.008	0.468	***	0.497	100%
Sweden	GBM	Imposed Weights	0.01	0.011	0.459	***	0.512	75%
United States	GBM	Unrestricted	0.008	0.007	0.569	***	0.772	75%

Note: Models include the best performing model according to the RMSE (Root Mean Squared Error) criterion. RRMSE: Relative Root Mean Squared Error (obtained as the ratio of each model RMSE with that of the DFM one). FDA: Forecast Directional Accuracy. Method corresponds to the MIDAS specification employed (See Annex B for more details). GBM: Gradient Boosting Trees (See Annex C for more details). Stars indicate the significance of the Diebold-Mariano (small sample) p-values as follows 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1. Tests include the Harvey-Leybourne-Newbold correction for small samples.

Annex F. Predicted value against observed productivity growth per country

Figure A F.1. Evolution of predicted labour productivity growth

Predicted (blue) against observed (red), Percentage



Note: Panel A and B split the sample in halves by alphabetical order. Blue (red) lines correspond to predicted (actual) productivity growth. Country-specific predictions are obtained from selected best models and MIDAS methods (See Section 2 for a complete description). Türkiye is excluded from the sample due to a large share of missing input data.
Source: Authors' calculations.